

# GEOMETRY BY CONSTRUCTION: A GEOMETRIC PRIOR BOTTLENECK

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## ABSTRACT

The conditional entropy bottleneck (CEB) compresses representations against a label-conditional reference whose geometry is left entirely to the optimizer. We show that this reference has two failure channels: in deterministic settings the CEB knob is dead (a known Caveat), and, more fundamentally, the certificate is blind to the relationships between labels: no cross-class term constrains the geometry among the class priors. We close this channel by declaring the geometry: the prior is confined to a qualified geometric family computed statelessly at every training step: an orthogonal class frame (OPB) for classification and an isometric regression axis (EPB) for regression, both with prediction rules tied to the geometry by construction (the zero-parameter anchor-energy classifier and the tied projection head). The framework preserves the CEB certificate for any stateless conditional prior and yields a comparator-dependent alignment bound for arbitrary label noise and model misspecification. Its residual floor separates irreducible label ambiguity, encoder approximation, and covariance mismatch; the exact  $O(1/\beta)$  rate is recovered only as the noiseless, exactly realizable special case. A controlled known-noise experiment recovers the predicted non-zero floors at three noise levels and the zero floor when labels are deterministic. Independently of this optimization bound, the observed anchor mismatch transfers the declared prior separation to the posterior. At the strongest reported compression, direct benchmark diagnostics show this bound is informative for 96.9% of ImageNet-100 class pairs with the largest anchors, while the regression bound is consistent but often vacuous. The construction removes the unspecified inter-class geometry of the reference. Across nine datasets and four backbones, GPB is best or tied-best in 7 of 12 settings and dominates CEB everywhere, and the anchor scale converts the  $\beta$ -knob from a collapse dial into a plateau dial.

## 1 INTRODUCTION

The information bottleneck (IB) principle (Tishby et al., 2000) seeks a representation  $Z$  of an input  $X$  that retains the information needed to predict the label  $Y$  while discarding the rest. The variational IB (VIB) (Alemi et al., 2017) bounds the compression term with a fixed marginal prior  $\mathcal{N}(0, I)$ ; the conditional entropy bottleneck (CEB) (Fischer, 2020) sharpens the bound to the residual information by replacing the marginal prior with a trainable label-conditional reference  $b(z|y)$ , certified by the gap  $\text{ReX} = \mathbb{E}[\text{KL}(q(z|x) \| b(z|y))] \geq I(X; Z|Y)$ . We call this gap the *certificate*; its failure modes are our starting point.

**The problem: a reference with no declared geometry.** Two facts about CEB are established. First, *the knob is dead*: in the deterministic setting  $Y = f(X)$ , every compression weight selects the same corner of the information plane (Kolchinsky & Tracey, 2019); this is settled background, repaired by their squared compression term, and we return to it only as a mechanism connection (Proposition A.4, Appendix). Second, and less recognized, *the certificate is blind to the relationships between labels*: the CEB compression term is a sum of per-class matching terms (each sample’s KL touches only its own class prior, and no term couples two distinct class priors), so at its class-wise matched optimum it reduces to  $I(X; Z|Y)$ , irrespective of the inter-class geometry realized by those matched conditionals (Proposition 2.1). Class means may merge, coincide, or point anywhere without changing the certificate; on the family  $T_\alpha = B_\alpha \cdot Y$  that interpolates from collapse to the exact copy of the label, the certificate reads zero across the whole retention axis (Proposi-

tion A.2, Appendix). CEB therefore compresses against a reference whose inter-class geometry the optimizer may choose at will: what a properly compressed representation should look like (class means separated, ordered, or laid out on a frame) is exactly what the certificate leaves unspecified.

**A geometric prior bottleneck.** We close this gap by *declaring* the geometry of the reference. The prior  $p(z|y)$  is confined to a *qualified geometric family*: label-conditional Gaussians with a bounded diagonal covariance and a mean map that encodes the label structure at a fixed, declared scale: a two-sided separation condition for discrete labels and a two-sided metric condition for continuous labels (Section 2.1). The prior is recomputed statelessly at every step, so the degenerate references — merged classes, a collapsed point, an arbitrarily rescaled geometry — are excluded by construction rather than by the optimizer’s good behavior. We call this family of objectives a *geometric prior bottleneck* (GPB). Two instantiations realize the principle: *OPB* (orthogonal prior bottleneck) for classification, where the prior means form the scaled orthogonal frame  $a q_{\theta,k}$  (every pair of distinct classes at distance  $\sqrt{2}a$ , with class-mean Gram matrix  $a^2 I_K$  by construction), and *EPB* (equidistant prior bottleneck) for regression, where the prior means lie on a unit axis,  $\rho \tilde{y} u$ , so latent distances equal  $\rho$  times standardized label distances exactly. Both keep the CEB posterior, a single KL, and the residual certificate unchanged; the classification head is the anchor-energy classifier  $\propto \exp(-\frac{1}{2\tau^2} \|z - a q_{\theta,k}\|^2)$  and the regression head the tied projection  $\hat{y} = u^\top z / \rho$ : prediction rules tied to the prior geometry by construction, with no free parameters. The constraints bind the *reference*, not the deployed representation; the framework constrains the geometric *form* of the label code and does not claim that the certificate meters retained label information.

**Theoretical guarantees and scope.** The theory has three levels. First, the certificate gap identity holds pointwise for any stateless prior (Proposition 2.3). Second, for arbitrary  $P_{XY}$  and any feasible comparator  $w_0$ , every  $\varepsilon$ -near-optimum obeys the oracle inequality

$$\mathbb{E}[D_{\hat{w}}(Y)] \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta},$$

where  $\mathcal{R}$  is prediction risk and  $\mathcal{K}$  is the conditional KL (Theorem 2.1). For OPB and EPB,  $\mathcal{K}(w_0)$  decomposes into an irreducible conditional-label-variability term, encoder approximation, and covariance mismatch. Thus noisy or capacity-limited problems have an explicit alignment floor rather than an unsupported convergence-to-zero claim. The familiar  $O(1/\beta)$  statement follows only as the zero-floor corollary: deterministic labels, exact posterior–prior matching, and a compatible prediction head (Corollary 2.1). Third, without any optimization or realizability premise, the realized mismatch transfers prior separation to posterior separation (Proposition 2.4). None of these statements guarantees task accuracy; they characterize the certificate and the transfer of declared geometry.

**Experiments.** We separate predictive performance from evidence about the theoretical mechanism. The twelve real-data settings compare utility: GPB is best or tied-best in 7 of 12 settings and dominates CEB in all 12, but these comparisons are not presented as verification of an alignment rate. The mechanism analysis instead asks whether the constructed prior geometry is transferred to the posterior and can be read out without a free prediction head. Direct mismatch measurements make this link quantitative: large OPB anchors retain positive separation bounds for most ImageNet-100 class pairs, whereas EPB transfers the regression axis more strongly than its scale. A separate controlled experiment spans deterministic and noisy labels: it recovers the analytic ambiguity floor throughout the  $\beta$  sweep, while making no claim that the real-data comparator floors are known.

Our contributions are: (1) the identification and formalization of the missing cross-label geometric constraint in CEB (Proposition 2.1); (2) the GPB framework and its two instances, OPB and EPB, with prediction rules tied to the prior geometry; (3) a comparator-dependent alignment bound for arbitrary label distributions and approximate model families, with explicit ambiguity, encoder, and covariance terms, plus the exact-realizability result as a zero-floor corollary; (4) unconditional certificate preservation and mismatch-dependent separation transfer for any qualified geometric prior; and (5) performance comparisons separated explicitly from direct mismatch diagnostics and a controlled known-noise illustration of the analytic ambiguity floor.

## 2 METHOD

### 2.1 FROM CEB TO GPB

Let  $X$  be the input and  $Y$  the label with  $K$  classes; the encoder induces the Markov chain  $Y - X - Z$ . Following Fischer (2020), the variational CEB objective is  $\min_{e,b,c} \langle \log e(z|x) \rangle - \langle \log b(z|y) \rangle - \gamma \langle \log c(y|z) \rangle$  over the stochastic encoder  $e$ , the backward encoder  $b$ , and the classifier  $c$ . We use the diagonal Gaussian posterior  $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$ , the loss convention  $\text{CE} + \beta \text{KL}$  ( $\beta = 1/\gamma$ ), and the deterministic code  $z = \mu_\phi(x)$  at evaluation. The central object is the residual certificate: for any conditional prior  $r(z|y)$ ,

$$\mathbb{E}[\text{KL}(q_\phi(z|x) \| r(z|y))] = I(X; Z|Y) + \mathbb{E}_y[\text{KL}(q_\phi(z|y) \| r(z|y))] \geq I(X; Z|Y), \quad (1)$$

by Fischer’s chain-rule identity. CEB instantiates  $r(z|y) = b_\theta(z|y)$ ; VIB fixes  $r = \mathcal{N}(0, I)$ . The certificate’s failure mode is that its compression term carries no cross-class structure:

**Proposition 2.1** (The certificate records nothing about inter-label geometry). *The CEB compression term contains no explicit term coupling two distinct class priors: it is a sum of per-class matching terms, each sample’s KL touching only its own class prior. Consequently, at its class-wise matched optimum ( $b = q(z|y)$ ), the certificate reduces to*

$$\text{ReX} = I(X; Z|Y), \quad (2)$$

*irrespective of the inter-class geometry realized by the matched conditionals: class means merged, coincident, or placed in arbitrary directions all yield the same certificate value. (The prediction term couples classes through the softmax; the statement is at a fixed encoder with a matched backward encoder.)*

At the matched optimum the certificate thus carries no information about the relationships among the class priors, and on the retention family  $T_\alpha = B_\alpha \cdot Y$  of Kolchinsky & Tracey (2019) it reads zero from total collapse to the full copy (Proposition A.2, Appendix). We close this channel by declaring the geometry of the reference.

Let  $\mathcal{G}$  be a *qualified geometric prior family*: a set of label-conditional distributions  $p_\theta(z|y)$  that depend on  $y$  alone, computed from the prior encoder, with a bounded diagonal covariance,

$$p_\theta(z|y) = \mathcal{N}(m_\theta(y), \Sigma_\theta(y)), \quad \Sigma_\theta(y) \text{ diagonal}, \quad \Sigma_\theta(y) \preceq \tau_{\max}^2 I_d \text{ for all } y, \quad (3)$$

and a mean map that encodes the label structure at a fixed, declared scale: for discrete labels a two-sided separation condition,

$$\Delta \leq \|m_\theta(y) - m_\theta(y')\| \leq \bar{\Delta} \text{ for all } y \neq y', \quad (4)$$

and for continuous labels with a declared label metric  $d_Y$ ,

$$\rho_{\min} d_Y(y, y') \leq \|m_\theta(y) - m_\theta(y')\| \leq \rho_{\max} d_Y(y, y'). \quad (5)$$

The declared constants  $\Delta, \bar{\Delta}, \rho_{\min}, \rho_{\max}$  are fixed before training and not trained. The lower bounds encode the label distinctions; the upper bounds close the scale escape: without them the family would contain  $m_\theta(y) = s m_0(y)$  for arbitrarily large  $s$ , and the objective might lack a finite optimum, making “geometry by construction” vacuous. The GPB objective is

$$\mathcal{L}_{\text{GPB}} = \mathbb{E}_{(x,y)} \left[ \text{CE}(c_\psi(z), y) + \beta \text{KL}(q_\phi(z|x) \| p_\theta(z|y)) \right], \quad z \sim q_\phi(z|x), \quad p_\theta \in \mathcal{G}. \quad (6)$$

### 2.2 GEOMETRIC INSTANTIATIONS

#### 2.2.1 OPB: AN ORTHOGONAL CLASS FRAME (CLASSIFICATION)

For classification, let  $K \leq d$ . The prior encoder  $g_\theta$  is a single unactivated linear map, evaluated on the all-class one-hot matrix  $E_K$  at every forward pass, giving the raw class-mean matrix

$$U_\theta = g_\theta(E_K)^\top \in \mathbb{R}^{d \times K}, \quad (7)$$

orthonormalized by the thin polar factor

$$Q_\theta = U_\theta (U_\theta^\top U_\theta)^{-1/2}, \quad Q_\theta^\top Q_\theta = I_K, \quad (8)$$

computed via the thin SVD  $U_\theta = W\Sigma V^\top$ ,  $Q_\theta = WV^\top$  (no sign convention needed). The polar factor is exactly equivariant under relabeling of the classes,  $Q_\theta(U_\theta P) = Q_\theta(U_\theta)P$  (derivation in Appendix A.3), so no class index plays a distinguished role. We make one standing assumption, verified, not imposed, in the implementation:

**Assumption 2.1** (Full rank).  $U_\theta$  has full column rank at every training step, so that Eq. equation 8 and its backward pass are defined. The condition holds almost surely at the identity initialization, and  $\sigma_{\min}(U_\theta)$  is monitored along training (Appendix A.3).

The geometric prior replaces the raw means by the scaled frame

$$p_\theta^{\text{OPB}}(z|y=k) = \mathcal{N}(a q_{\theta,k}, \tau^2 I_d), \quad q_{\theta,k} := Q_\theta[:, k], \quad (9)$$

with anchor scale  $a > 0$  and fixed shared prior variance  $\tau^2$  (default 1.0). This canonical instance, used by the theory controls, removes the covariance channel entirely; the covariance-relaxed real-data protocol is specified below. Since  $Q_\theta^\top Q_\theta = I_K$ , the class-mean Gram matrix is  $a^2 I_K$  by construction:

$$a_i^\top a_j = a^2 \delta_{ij}, \quad \|a_k\| = a, \quad \|a_i - a_j\|^2 = 2a^2 \quad (i \neq j). \quad (10)$$

The frame is stateless: an immediate function of the current prior-encoder parameters, computed from the all-class table: no EMA, no prototypes, no batch statistics. The single-KL objective is

$$\mathcal{L}_{\text{OPB}} = \mathbb{E}_{(x,y)} \left[ \text{CE}(c_\psi(z), y) + \beta \text{KL}(q_\phi(z|x) \| p_\theta^{\text{OPB}}(z|y)) \right], \quad z \sim q_\phi(z|x). \quad (11)$$

The main classifier is the anchor-energy classifier, whose scale is fixed by construction:

**Proposition 2.2** (Anchor-energy classifier equivalence). *The scores*

$$s_k(z) = -\frac{\|z - a q_{\theta,k}\|^2}{2\tau^2} + \log \pi_k \quad (12)$$

are the class- $k$  Gaussian log posterior up to a class-independent term; for uniform classes the equivalent linear classifier has weights  $w_k = (a/\tau^2) q_{\theta,k}$ : equal norms and pairwise orthogonal rows. The prediction rule therefore uses exactly the same geometry as the prior, with no free classifier scale (proof in Appendix A.3).

The posterior logvar head is zero-initialized and the prior encoder identity-initialized ( $U_\theta = [e_1, \dots, e_K]$ , full column rank), so training starts at  $\text{KL} \approx a^2/(2\tau^2)$  with a stable polar backward. The plain linear classifier is kept in the experiments as a free-head variant; it does not enjoy the fixed-scale condition of Proposition A.4.

### 2.2.2 EPB: AN EQUIDISTANT REGRESSION AXIS (REGRESSION)

For a one-dimensional continuous label, standardize once on the training set,  $\tilde{y} = (y - y_{\text{center}})/y_{\text{scale}}$ , with the statistics frozen for validation and test. Let  $W \in \mathbb{R}^{d \times 1}$  be a trainable direction and  $u = W/\|W\|$  its per-step normalization. The isometric prior is

$$p^{\text{EPB}}(z|y) = \mathcal{N}(\rho \tilde{y} u, \tau^2 I_d), \quad \|\mu_p(y_i) - \mu_p(y_j)\| = \rho |\tilde{y}_i - \tilde{y}_j|, \quad (13)$$

with isometric scale  $\rho > 0$  and fixed shared prior variance  $\tau^2$  in the canonical instance: latent distances equal  $\rho$  times standardized label distances exactly, at every scale. Order and magnitude of label differences are encoded with equality, which qualifies the family (Eq. equation 3 with  $\tau_{\max}^2 = \tau^2$  and Eq. equation 5 with  $\rho_{\min} = \rho_{\max} = \rho$  on the standardized label metric). The main variant uses the trainable normalized axis and no bias (the fixed-axis and unconstrained-scale ablations are in Appendix A.3). The prediction head is the *tied projection*  $\hat{y} = u^\top z/\rho$ , the regression analogue of the anchor-energy classifier: geometrically consistent, with no free direction or scale that could bypass the prior; the ordinary regression head is kept as a free-head variant. The objective is  $L_{\text{reg}}(\hat{y}, y) + \beta \text{KL}(q_\phi(z|x) \| p^{\text{EPB}}(z|y))$ : a single KL, stateless.

**Covariance protocol.** Equations equation 9 and equation 13 define the fixed-covariance canonical instances used for the exact Bayes-head equivalence and the controlled label-ambiguity experiment. The real-data benchmark protocol retains the common CEB parameterization of a learned label-conditional diagonal covariance, with log-variance clamped to a fixed interval. It is therefore a

covariance-relaxed member of  $\mathcal{G}$ : the OPB/EPB mean geometry and readout scale remain fixed, while Eq. equation 3 holds with the clamp-induced  $\tau_{\max}^2$ . Proposition 2.2 is an exact Gaussian Bayes equivalence only for the shared-covariance canonical instance; in the relaxed benchmark it remains a zero-parameter readout of the same anchor means. The direct diagnostics in Section 3.8.1 use each checkpoint’s realized conditional covariance rather than silently substituting the canonical  $\tau^2 I$ .

### 2.3 ALIGNMENT UNDER APPROXIMATE REALIZABILITY

The theory separates three statements with different scopes. The certificate identity holds for every stateless conditional prior. The geometric separation transfer holds for the mismatch realized by any fitted model. The optimization-dependent statement is an oracle inequality against any feasible comparator in the actual model and head families; it allows stochastic labels and approximate rather than exact posterior–prior matching. Only its zero-approximation-error special case has a pure  $O(1/\beta)$  rate to zero.

The first, unconditional link holds pointwise:

**Proposition 2.3** (A stateless geometric prior preserves the CEB certificate). *For any prior family  $\mathcal{G}$  be qualified with variance bound  $\tau_{\max}^2$ . (a) Discrete labels. With posterior class centers  $m_k := \mathbb{E}[\mu_\phi(X)|Y = k]$  and class-wise anchor mismatch*

$$\mathbb{E}[\text{KL}(q_\phi(z|x) \parallel p_\theta(z|y))] = I_{q_\phi}(X; Z|Y) + \mathbb{E}_Y[\text{KL}(q_\phi(z|Y) \parallel p_\theta(z|Y))] \geq I_{q_\phi}(X; Z|Y). \quad (14)$$

A batch-dependent prior would break the conditional-prior argument.

**Proposition 2.4** (Mismatch-dependent separation transfer for any qualified geometric prior). *Let  $\mathcal{G}$  be qualified with variance bound  $\tau_{\max}^2$ . (a) Discrete labels. With posterior class centers  $m_k := \mathbb{E}[\mu_\phi(X)|Y = k]$  and class-wise anchor mismatch*

$$D_k := \frac{1}{2\tau_{\max}^2} \mathbb{E}[\|\mu_\phi(X) - m_\theta(k)\|^2 \mid Y = k], \quad (15)$$

for every pair  $i \neq j$ ,

$$\|m_i - m_j\| \geq \|m_\theta(i) - m_\theta(j)\| - \sqrt{2\tau_{\max}^2 D_i} - \sqrt{2\tau_{\max}^2 D_j} \geq \Delta - \sqrt{2\tau_{\max}^2 D_i} - \sqrt{2\tau_{\max}^2 D_j}, \quad (16)$$

using Eq. equation 4. (b) Continuous labels. For Borel-measurable  $X, Y$  with integrable  $\mu_\phi(X)$ , the regular conditional posterior center  $m(y) := \mathbb{E}[\mu_\phi(X) \mid Y = y]$  and pointwise mismatch

$$D(y) := \frac{1}{2\tau_{\max}^2} \mathbb{E}[\|\mu_\phi(X) - m_\theta(y)\|^2 \mid Y = y] \quad (17)$$

are defined for  $P_Y$ -a.e.  $y$ , and for  $P_Y$ -a.e. pair  $y \neq y'$ ,

$$\begin{aligned} \|m(y) - m(y')\| &\geq \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')} \\ &\geq \rho_{\min} d_Y(y, y') - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')}. \end{aligned} \quad (18)$$

We next state the optimization link without assuming deterministic labels or an exactly aligned posterior. Let  $w = (\phi, \theta, \psi)$  denote all fitted parameters and define

$$\mathcal{R}(w) := \mathbb{E}[\ell(c_\psi(Z), Y)], \quad \mathcal{K}(w) := \mathbb{E}[\text{KL}(q_\phi(z|X) \parallel p_\theta(z|Y))], \quad (19)$$

where  $\ell \geq 0$ , and

$$\mathcal{D}(w) := \frac{1}{2\tau_{\max}^2} \mathbb{E}[\|\mu_\phi(X) - m_\theta(Y)\|^2]. \quad (20)$$

**Theorem 2.1** (Comparator-dependent alignment under approximate realizability). *Let  $\hat{w}$  satisfy  $\mathcal{R}(\hat{w}) + \beta\mathcal{K}(\hat{w}) \leq \inf_w \{\mathcal{R}(w) + \beta\mathcal{K}(w)\} + \varepsilon$ . For every feasible comparator  $w_0$  in the same encoder, prior, posterior, and head families,*

$$\mathcal{D}(\hat{w}) \leq \mathcal{K}(\hat{w}) \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta}. \quad (21)$$

Consequently, the existence of a comparator with  $\mathcal{K}(w_0) \leq \delta_{\text{app}}$  and  $\mathcal{R}(w_0) \leq C_{\text{app}}$  implies

$$\mathcal{D}(\hat{w}) \leq \delta_{\text{app}} + \frac{C_{\text{app}} + \varepsilon}{\beta}. \quad (22)$$

No assumption on whether  $Y$  is deterministic is needed. Approximation error and irreducible label ambiguity enter through the comparator  $\text{KL } \delta_{\text{app}}$ , while optimization error enters through  $\varepsilon/\beta$ .

For the two concrete GPB instances, the approximation floor has an explicit interpretation. Let

$$V_0 := \frac{1}{2} \mathbb{E} \left[ \sum_{r=1}^d \left( \frac{\sigma_{0,r}^2(X)}{\tau^2} - 1 - \log \frac{\sigma_{0,r}^2(X)}{\tau^2} \right) \right] \quad (23)$$

be the comparator's covariance mismatch against the shared variance  $\tau^2 I_d$ .

**Proposition 2.5** (Ambiguity–approximation decomposition). (a) OPB classification. Write  $\eta_k(x) := P(Y = k \mid X = x)$  and consider any feasible comparator with posterior  $q_0(z|x) = \mathcal{N}(h(x), \text{diag}(\sigma_0^2(x)))$  and OPB frame  $\{q_{0,k}\}_{k=1}^K$ . Its KL is exactly

$$\mathcal{K}(w_0) = \underbrace{\frac{a^2}{2\tau^2} \mathbb{E}[1 - \|\eta(X)\|_2^2]}_{\text{irreducible label ambiguity}} + \underbrace{\frac{1}{2\tau^2} \mathbb{E} \left[ \left\| h(X) - a \sum_k \eta_k(X) q_{0,k} \right\|^2 \right]}_{\text{encoder mean approximation}} + \underbrace{V_0}_{\text{covariance mismatch}}. \quad (24)$$

(b) EPB regression. Suppose  $\mathbb{E}[\tilde{Y}^2] < \infty$ , write  $\nu(x) := \mathbb{E}[\tilde{Y} \mid X = x]$ , and consider a feasible comparator with mean  $h(x)$  and axis  $u_0$ . Then

$$\mathcal{K}(w_0) = \underbrace{\frac{\rho^2}{2\tau^2} \mathbb{E}[\text{Var}(\tilde{Y} \mid X)]}_{\text{irreducible label ambiguity}} + \underbrace{\frac{1}{2\tau^2} \mathbb{E}[\|h(X) - \rho\nu(X)u_0\|^2]}_{\text{encoder mean approximation}} + \underbrace{V_0}_{\text{covariance mismatch}}. \quad (25)$$

Using either displayed quantity as  $\delta_{\text{app}}$  in Eq. equation 22 gives a non-zero, problem-dependent floor.

The original exact-alignment statement is now a special case rather than the principal guarantee:

**Corollary 2.1** (Zero-floor exact-realizability special cases). (a) Classification. Assume  $Y = f(X)$ , the posterior can realize  $q(z|x) = p_\theta(z|y)$ , and the classifier family contains the aligned mixture's Bayes posterior

$$c_k(z) = \frac{\pi_k |\Sigma_\theta(k)|^{-1/2} e^{-\frac{1}{2}(z - m_\theta(k))^\top \Sigma_\theta(k)^{-1} (z - m_\theta(k))}}{\sum_j \pi_j |\Sigma_\theta(j)|^{-1/2} e^{-\frac{1}{2}(z - m_\theta(j))^\top \Sigma_\theta(j)^{-1} (z - m_\theta(j))}}. \quad (26)$$

Then  $\mathcal{K}(w_0) = 0$ ,  $C_G := \mathcal{R}(w_0) = H(Y \mid Z_{\text{align}}) \leq \ln K$ , and

$$\mathcal{D}(\hat{w}) \leq \frac{C_G + \varepsilon}{\beta} \leq \frac{\ln K + \varepsilon}{\beta}. \quad (27)$$

For shared-covariance OPB, Eq. equation 26 is exactly the anchor-energy classifier. (b) Regression. If  $Y = f(X)$ , the posterior can realize the EPB prior, and the tied projection head is used with standardized squared loss, then  $\mathcal{K}(w_0) = 0$ ,  $\mathcal{R}(w_0) = \tau^2/\rho^2$ , and

$$\mathcal{D}(\hat{w}) \leq \frac{\tau^2/\rho^2 + \varepsilon}{\beta}. \quad (28)$$

**Corollary 2.2** (Comparator-dependent separation at near-optimal solutions). Fix any comparator  $w_0$  in Theorem 2.1 and write  $B_0 := \mathcal{K}(w_0) + (\mathcal{R}(w_0) + \varepsilon)/\beta$ . (a) Classification. For classes  $i \neq j$  with  $\pi_i, \pi_j > 0$ ,

$$\|m_i - m_j\| \geq \|m_\theta(i) - m_\theta(j)\| - \sqrt{\frac{2\tau_{\max}^2 B_0}{\pi_i}} - \sqrt{\frac{2\tau_{\max}^2 B_0}{\pi_j}}. \quad (29)$$

(b) Regression. For any  $t > 0$ , the label set  $S_t := \{y : D(y) \leq t\}$  has  $P_Y$ -measure at least  $1 - B_0/t$ , and the pointwise transfer of Proposition 2.4(b) yields, for  $P_Y$ -a.e.  $y, y' \in S_t$ ,

$$\|m(y) - m(y')\| \geq \rho_{\min} d_Y(y, y') - 2\sqrt{2\tau_{\max}^2} t; \quad (30)$$

for EPB ( $\rho_{\min} = \rho$ ,  $\tau_{\max} = \tau$ ) the bound reads  $\rho d_Y(y, y') - 2\sqrt{2\tau^2} t$ .

The orthogonal instance upgrades these statements with a constructive fact:

**Proposition 2.6** (Positive rate cost of class-independent means). *If the posterior mean is class-independent,  $\mathbb{E}[\mu_\phi(X)|Y = k] = c$  for every  $k$ , the mean-mismatch part of the rate satisfies*

$$\mathbb{E}\left[\frac{1}{2\tau^2}\|\mu_\phi(X) - a q_{\theta,Y}\|^2\right] \geq \frac{a^2}{2\tau^2}\left(1 - \sum_{k=1}^K \pi_k^2\right), \quad (31)$$

which for uniform classes reads  $a^2(1 - 1/K)/(2\tau^2)$ : the VIB corner  $\mu \equiv 0$  is a penalized configuration rather than a free optimum, whenever  $a > 0$  and two classes have positive probability.

**Corollary 2.3** (The orthogonal instance instantiates the conditional separation transfer). *The orthogonal instance is qualified with  $\tau_{\max}^2 = \tau^2$  and  $\Delta = \bar{\Delta} = \sqrt{2}a$ ; Proposition 2.4 specializes to*

$$\|m_i - m_j\| \geq \sqrt{2}a - \sqrt{2\tau^2 D_i} - \sqrt{2\tau^2 D_j}, \quad D_k := \frac{1}{2\tau^2} \mathbb{E}\left[\|\mu_\phi(X) - a q_{\theta,k}\|^2 \mid Y = k\right]. \quad (32)$$

*Scope.* The geometric constraints bind the reference, not the deployed representation: at any frame rotation there exists a full-retention label code with a zero certificate (Remark A.3, Appendix), so the framework constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. The proofs, the dead-knob connection (the anchored reference supplies the quadratic displacement behind the strict convexity of Proposition A.4), and the implementation details are in the appendix.

### 3 EXPERIMENTS

The experiments are organized into three evidence tracks. The first is a performance comparison: does replacing CEB’s unconstrained reference by a constructed geometry improve end-to-end prediction, and does prediction survive when the readout is tied to that geometry? The main sweep answers these questions on twelve dataset–backbone settings, with compression–accuracy curves in Section 3.5 and adversarial robustness in Section 3.7. The second is mechanism analysis: does the declared geometry actually appear in the fitted posterior? Section 3.8 uses geometric diagnostics and directly analyzes the mismatch appearing in Proposition 2.4. The third is controlled theory illustration: Section 3.2 varies known label ambiguity continuously from the exactly realizable zero-floor case to non-zero analytic floors. Performance improvements in the first track are not treated as verification of the alignment bounds in the other two.

#### 3.1 EXPERIMENTAL SETUP

**Datasets.** Twelve settings over nine datasets and four backbones, spanning image, graph, and text modalities. *Classification:* MNIST (MLP and CNN backbones); ImageNet-100 (MLP and CNN) on frozen ResNet-50 features (layer-3 activations pooled to  $4 \times 4$ ); Cora, transductive node classification (GCN); IMDB sentiment (RNN); AG News (RNN) on frozen BERT layer-10 token features reduced by PCA to 50 dimensions. *Regression:* California Housing (MLP); STS-B (RNN with frozen GloVe-100d embeddings); ZINC-12k molecular graphs (GCN, the official 10k/1k/1k split); AgeDB face age (MLP and CNN). Where no official split exists we use a fixed 60/20/20 train/validation/test partition (seed 42); on MNIST 10k of the 60k training images are held out for validation. Regression targets are standardized to  $[0, 1]$  on the training set (the test metric is re-normalized).

**Backbones and protocol.** The MLP uses hidden layers 512–256; the CNN two convolutional blocks (channels 32, 64) followed by a 256-dim hidden layer; the GCN matches the MLP sizes; the RNN is a two-layer LSTM with 512 and 256 units (a single 256-unit layer with max time-pooling on STS-B). All methods share the backbone, dropout 0.2, and bottleneck dimension  $d = 256$ . Every run trains with Adam (learning rate  $10^{-3}$ ;  $10^{-2}$  on Cora), at most 100 epochs with early stopping on the validation metric (patience 10), batch size 512 (128 on ZINC), and evaluates the checkpoint with the best validation score on the test set. GPB and GPB-L train five seeds per configuration (seeds 0–4); each variational baseline trains fifteen seeds per  $\beta$  value (seeds 0–14), so that every method trains a matched total budget ( $6 \times 3 \times 5 = 6 \times 15 = 90$  runs per method). The tables report the mean over the runs. Classification early-stops on validation macro-AUC and reports test accuracy; regression early-stops on validation  $R^2$  and reports test  $R^2$ .

**Baselines.** The plain backbone (Base); VIB (Alemi et al., 2017) with the fixed  $\mathcal{N}(0, I)$  prior; SVIB, the squared compression term of Kolchinsky & Tracey (2019) ( $\text{CE} + \beta \text{KL}^2$ ); NIB (Kolchinsky et al., 2019), the noise-injection bottleneck with the pairwise-distance upper bound on  $I(X; M)$ ; CEB (Fischer, 2020) with the trainable label-conditional backward encoder; and the supervised  $\beta$ -DVCCA of Abdelaleem et al. (2025), which augments the VIB loss with an input reconstruction term,  $\text{CE} + \beta (\text{KL} + \text{MSE}_{\text{recon}})$ . All variational methods share the encoder, the posterior heads, and the loss convention of Eq. equation 6.

**External baselines.** Two non-IB comparison methods complete the reference set. *NCBD* (Koromilas et al., 2026) (Neural Collapse by Design) normalizes features and trainable class prototypes to the unit hypersphere and replaces the classification loss by the NONL contrastive objective, whose global optimum is the simplex ETF; its temperature  $\tau$  occupies the  $\beta$  slot of the sweep (grid  $\{0.05, 0.1, 0.2, 0.3, 0.5, 1\}$ ), and it applies to classification settings only. *AdaCap* (Belucci et al., 2026) targets small-sample regression: a permutation-based contrastive loss plus a Tikhonov closed-form output layer  $\hat{y} = H (H^\top H + \lambda I)^{-1} H^\top y$  with learnable  $\lambda$ , initialized from the swept grid  $\{10^{-2}, \dots, 10^3\}$ ; at evaluation the readout is refit on the training-set representations. It applies to regression settings only. Both train under the shared backbones, splits, early-stopping protocol, and five seeds per configuration.

**Our variants.** *GPB* denotes our method in the tables. On classification rows it is *OPB*, the orthogonal-frame prior of Eq. equation 9 with the anchor-energy classifier of Eq. equation 12: the prediction rule has no free parameters, its means and scale are tied to the prior mean geometry, and  $K \leq d$  holds in all settings (max  $K = 100$  on ImageNet-100); on regression rows it is *EPB*, the isometric prior of Eq. equation 13 with the tied projection head  $\hat{y} = u^\top z / \rho$ . Both use the covariance-relaxed protocol on the real-data benchmarks. *GPB-L* is the free-head ablation: the identical prior and KL, with an ordinary trainable linear classifier (or regression head) on  $z$ , so prediction is free to bypass the constructed geometry. The frame is orthonormalized from the all-class mean matrix at every forward pass by the thin SVD polar factor of Eq. equation 8 ( $Q_\theta = W V^\top$ ); the prior encoder is identity-initialized (initial anchors  $e_1, \dots, e_K$ , initial  $\text{KL} \approx a^2/2$  under unit variances), the variance heads are zero-initialized, and the construction is stateless: no EMA, no prototypes, no batch statistics. Six attribution variants (NCM-learn/NCM-ortho, CEB-energy/CEB-tied, OPB-FS/EPB-FS) switch the geometry, scale, readout, and IB factors separately; they are defined and evaluated in Section 3.4.

**Hyperparameter selection.** Each method selects its hyperparameters on the validation set from a shared grid: a six-value log-spaced  $\beta$  grid  $\{10^{-4}, \dots, 10\}$  for every variational baseline, and, for GPB and GPB-L, an additional three-value anchor-scale grid  $\{1, 6, 12\}$  ( $6 \times 3$  combinations). The main table reports, per method and setting, the test metric of the best validation configuration, so all methods are compared under an identical selection protocol (Base has no sweep and early-stops as usual).

**Separation of performance and mechanism evidence.** The main benchmark table is a performance comparison, not a test of an alignment theorem. The general comparator inequality, Theorem 2.1, applies without deterministic labels, but its benchmark-specific floor depends on unknown label ambiguity and model approximation terms. We therefore do not infer an alignment rate from test accuracy. The mechanism analysis separately measures whether the prior geometry reaches the posterior and whether its tied readout remains predictive. Finally, the controlled synthetic experiment isolates both theoretical regimes: exact realizability for the zero-floor corollary and known label noise for the non-zero floor of Proposition 2.5.

### 3.2 CONTROLLED LABEL AMBIGUITY: FROM THE ZERO TO THE NON-ZERO FLOOR

We isolate the ambiguity term of Proposition 2.5 in a distribution where it is known exactly. Let the clean class  $C$  be uniform on  $\{1, \dots, K\}$ , with  $K = 10$  and  $X = e_C \in \mathbb{R}^{10}$ . The observed label follows symmetric corruption,

$$P(Y = C \mid C) = 1 - r, \quad P(Y = j \mid C) = \frac{r}{K - 1} \quad (j \neq C), \quad (33)$$



| $r$ | $\delta_{\text{noise}}$ | $\widehat{D}(\beta = 10)$ | $\widehat{D} - \delta_{\text{noise}}$ | $\widehat{L} - L_{\text{comp}}$ |
|-----|-------------------------|---------------------------|---------------------------------------|---------------------------------|
| 0   | 0.000                   | $0.0000 \pm 0.0000$       | $+0.0000$                             | $0.000 \pm 0.000$               |
| 0.1 | 3.400                   | $3.4018 \pm 0.0000$       | $+0.0018$                             | $-0.005 \pm 0.004$              |
| 0.2 | 6.400                   | $6.4067 \pm 0.0002$       | $+0.0067$                             | $-0.053 \pm 0.009$              |
| 0.4 | 11.200                  | $11.2172 \pm 0.0003$      | $+0.0172$                             | $-0.191 \pm 0.018$              |

Table 1: Controlled label-ambiguity experiment. Means  $\pm$  standard deviations are over five seeds. The analytic floor is Eq. equation 34; the last column checks whether the fitted population objective improves on the explicit comparator.

for  $r \in \{0, 0.1, 0.2, 0.4\}$ . The encoder observes  $X$  but not  $Y$ . We use OPB with the fixed frame  $Q = I_K$ ,  $a = 6$ ,  $\tau = 1$ , the anchor-energy head, nine  $\beta$  values from  $10^{-3}$  to 10, and five seeds (180 main-grid fits). Training evaluates the population objective by weighting all  $K^2$  clean/observed-label pairs by Eq. equation 33; a descending- $\beta$  warm start is used to remove avoidable optimization error at the small- $\beta$  end. Thus  $r = 0$  is the zero-floor exact-realizability case, while  $r > 0$  violates  $Y = f(X)$  in a controlled way.

For  $\eta_k(x) = P(Y = k \mid X = x)$ , the analytic comparator has mean  $h_0(x) = a \sum_k \eta_k(x) q_k$  and covariance  $\tau^2 I$ . Its KL is exactly

$$\delta_{\text{noise}}(r) = \frac{a^2}{2\tau^2} \left[ 1 - (1 - r)^2 - \frac{r^2}{K - 1} \right], \quad (34)$$

which is 0, 3.4, 6.4, and 11.2 nats at the four noise levels. We also estimate the same comparator’s anchor-energy risk  $C_{\text{comp}}(r)$ . Whenever the fitted population objective satisfies  $\widehat{L} \leq L_{\text{comp}} = C_{\text{comp}} + \beta \delta_{\text{noise}}$ , non-negativity of the fitted risk gives the directly checkable empirical comparison

$$\widehat{D} \leq \widehat{\mathcal{K}} \leq \delta_{\text{noise}} + \frac{C_{\text{comp}}}{\beta}. \quad (35)$$

Table 1 shows the predicted transition from a zero to a non-zero floor. At  $r = 0$ ,  $\widehat{D}$  is zero to the stored precision over the complete  $\beta$  grid. For  $r > 0$ , it approaches the corresponding analytic floor; at  $\beta = 10$  the excess is only 0.0018, 0.0067, and 0.0172 nats. Across all population-grid points the excess remains below 0.032 nats, and every fitted objective satisfies  $\widehat{L} \leq L_{\text{comp}}$  at the reported Monte Carlo precision, so Eq. equation 35 holds at every reported point. The excess changes smoothly with  $\beta$ , but the floor dominates the entire sweep; we therefore do not interpret the curves as evidence of an exact  $1/\beta$  law. The tied readout’s clean-class and corrupted-label accuracies remain at their respective ceilings across the sweep, showing that the mismatch floor reflects irreducible label ambiguity rather than loss of the underlying class signal.

Fixed-posterior-variance and trainable-polar-frame controls reproduce the same analytic floors (Appendix A.5). A sampled-label version exposes the expected finite-sample and optimization sensitivity: six large- $\beta$  configurations fail the comparator-objective gate and are marked rather than counted as counterexamples. This experiment is a controlled illustration of the ambiguity term and the comparator inequality. It neither estimates the comparator floor on a real dataset nor turns the benchmark accuracy comparison into a theorem test.

### 3.3 MAIN RESULTS

Table 2 reports test accuracy (classification) and test  $R^2$  (regression), averaged over runs, for each method’s best configuration; bold marks the best mean per row and underline the second best, and the vertical rule separates GPB and its free-head ablation GPB-L from the baselines. NCB (classification settings only) and AdaCap (regression settings only) are external non-IB baselines under the same protocol; dashes mark the settings where a method does not apply. Statistical claims (best vs. tied) follow the paired-difference confidence intervals of Appendix A.8. Standard deviations are omitted for width and are reported in Appendix A.7. Three observations.

*The constructed geometry is free, and it usually pays.* GPB is best or tied-best in 7 of 12 settings with mean rank 1.83, against 4.83 for the plain backbone and 5.50–7.00 for the variational baselines, none of which is ever best; among the external baselines NCB never wins and AdaCap wins once

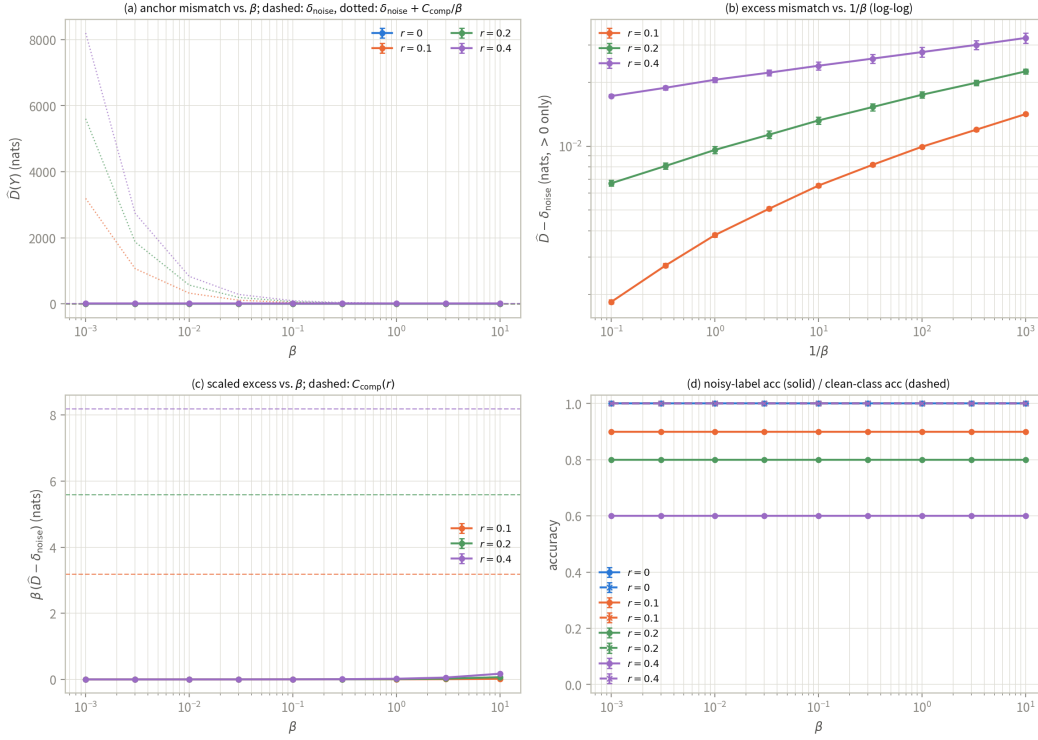


Figure 1: Controlled illustration of approximate realizability. (a) Anchor mismatch  $\hat{D}$  against  $\beta$ ; dashed lines are the analytic ambiguity floors and dotted lines the comparator references  $\delta_{\text{noise}} + C_{\text{comp}}/\beta$ . (b) Excess mismatch against  $1/\beta$ . (c) Scaled excess  $\beta(\hat{D} - \delta_{\text{noise}})$  against the comparator risks. (d) Accuracy on the corrupted label (solid) and clean class (dashed). Points are means and error bars are standard deviations over five seeds.

Table 2: Test accuracy (%) on classification tasks and test  $R^2$  on regression tasks (mean over runs). Each entry reports the best configuration over the tuned  $\beta$  (and anchor-scale) grid; the best entry per dataset is bolded. NCBD (classification settings only) and AdaCap (regression settings only) are external non-IB baselines; dashes mark the settings where a method does not apply.

| Dataset           | Backbone | Base          | VIB           | SVIB   | NIB    | CEB          | DVCCA  | NCBD  | AdaCap        | GPB           | GPB-L         |
|-------------------|----------|---------------|---------------|--------|--------|--------------|--------|-------|---------------|---------------|---------------|
| MNIST             | MLP      | 98.33         | 98.28         | 98.15  | 98.27  | <u>98.43</u> | 98.24  | 97.95 | —             | <b>98.54</b>  | 98.43         |
| MNIST             | CNN      | 99.13         | 99.28         | 99.23  | 99.22  | <u>99.30</u> | 99.29  | 99.18 | —             | <b>99.32</b>  | 99.31         |
| ImageNet-100      | MLP      | 83.12         | 82.84         | 83.02  | 82.35  | 82.77        | 82.99  | 83.64 | —             | <b>84.40</b>  | 84.01         |
| ImageNet-100      | CNN      | 83.40         | 82.94         | 83.11  | 82.91  | 82.97        | 83.21  | 80.56 | —             | <b>86.08</b>  | 84.88         |
| Cora              | GCN      | 87.68         | 86.83         | 86.46  | 86.35  | 86.27        | 86.90  | 87.34 | —             | <b>88.12</b>  | 87.86         |
| IMDb              | RNN      | 84.85         | 85.50         | 85.99  | 86.14  | 85.55        | 85.69  | 86.69 | —             | <u>86.84</u>  | <b>86.99</b>  |
| AG News           | RNN      | <b>92.94</b>  | 92.72         | 92.86  | 92.91  | 92.83        | 92.78  | 92.37 | —             | 92.87         | <u>92.92</u>  |
| Cal. Housing      | MLP      | 0.7618        | <u>0.7901</u> | 0.7897 | 0.7862 | 0.7894       | 0.7878 | —     | 0.7758        | 0.7895        | <b>0.7914</b> |
| STS-B             | RNN      | <b>0.2899</b> | 0.2719        | 0.2697 | 0.2417 | 0.2668       | 0.2713 | —     | 0.1613        | 0.2762        | 0.2755        |
| ZINC              | GCN      | <u>0.6348</u> | 0.6339        | 0.6325 | 0.6178 | 0.6257       | 0.6294 | —     | <b>0.6770</b> | 0.6344        | 0.6323        |
| AgeDB             | MLP      | 0.2180        | 0.3173        | 0.3385 | 0.2710 | 0.3317       | 0.3180 | —     | <u>0.3482</u> | <b>0.3536</b> | 0.3320        |
| AgeDB             | CNN      | 0.3914        | 0.3342        | 0.3323 | 0.3337 | 0.3882       | 0.3205 | —     | <u>0.4684</u> | <b>0.4700</b> | 0.4388        |
| Mean rank         |          | 4.83          | 5.75          | 5.50   | 7.00   | 5.96         | 6.00   | 6.43  | 4.40          | 1.83          | 2.54          |
| Best or tied-best |          | 2             | 0             | 0      | 0      | 0            | 0      | 0     | 1             | 7             | 2             |

(ZINC). By the paired bootstrap 95% CIs (Appendix A.8), the GPB–CEB difference is significantly positive in 8 of 12 settings and within the non-inferiority margin in three more; only on STS-B does the interval overlap zero beyond the margin  $([-0.009, +0.025])$ , so no setting shows a significant CEB advantage. GPB dominates CEB by mean in all 12 settings and is best among all methods on seven rows. The gains concentrate where the label structure is rich or supervision is scarce: +3.1

accuracy over CEB on ImageNet-100 with the CNN backbone (86.1 vs. 83.0), +1.9 on Cora, and +0.08  $R^2$  on AgeDB over the plain CNN (0.470 vs. 0.388). On saturated settings (MNIST) the margin is small but consistent, and on the two rows where the plain backbone remains best (STS-B, AG News) the gap is marginal ( $\leq 0.03$ ): the geometry is neutral there rather than harmful. On ZINC every bottleneck method trails the plain backbone, but the closed-form AdaCap readout edges it out (0.677 vs. 0.635  $R^2$ ), the one setting where any comparison method significantly beats GPB (see below).

*Prediction does not require the free head.* GPB-L, the free linear head on the same geometric prior, is best or tied-best in 2 of 12 settings (mean rank 2.54), and the GPB-GPB-L gap is within 0.4 accuracy or  $R^2$  points in 11 of 12 settings (the exception is ImageNet-100 CNN, 1.2 points): the zero-parameter anchor-energy classifier — whose weights are the frame itself — reaches the accuracy of a freely trained head in almost all settings, so the degrees of freedom removed by construction are not required for prediction, as claimed in the introduction. The three settings where GPB-L edges out GPB by mean (IMDb, AG News, Cal. Housing) do so with paired CIs overlapping zero, confirming that the tie does not inflate the numbers. On the five regression rows the tied projection head of EPB wins or matches the free regression head in four (the exception is Cal. Housing by 0.002).

*The tie regularizes the high-cardinality and low-data regimes.* The two settings with the largest GPB margins (ImageNet-100, with  $K = 100$ , and AgeDB, a small, high-variance regression task) are exactly those where an unconstrained classifier can overfit its own scale or the scarce supervision: fixing the prediction rule to the prior’s geometry removes both escape routes, consistent with the fixed-scale condition under which the aligned surrogate of Proposition A.4 has a unique interior optimum per  $\beta$ .

*External baselines do not close the gap.* By the paired CIs (Appendix A.8), GPB beats NCBP significantly on 6 of 7 classification settings; the exception is IMDb (+0.2 points, CI  $[-0.5, +0.7]$  crossing zero), where the GPB margin is within noise. The largest margin is ImageNet-100 with the CNN backbone (+5.5 points, CI  $[+4.3, +6.9]$ ). On the regression side GPB beats AdaCap significantly on Cal. Housing (+0.014  $R^2$ ) and STS-B (+0.115  $R^2$ ), and the AgeDB intervals straddle zero (the CNN one within the non-inferiority margin). The single reversal is ZINC, where AdaCap is the only method with a significant advantage over GPB ( $-0.043 R^2$ , CI  $[-0.046, -0.041]$ ): its ridge readout, refit on the training set at evaluation time, exploits the low-dimensional regression structure of molecular-graph labels without any compression objective. Neither external baseline performs information-bottleneck compression, so their sweeps exercise the same data but a different axis of the comparison.

### 3.4 ATTRIBUTION ABLATIONS: WHAT DRIVES THE GAIN?

The GPB construction bundles three mechanisms — the orthogonal/isometric geometry, the fixed anchor scale, and the tied geometric readout — with the IB objective itself, and the main table alone cannot attribute the gain among them. This section switches each factor off separately. All variants are trained under the protocol of Section 3.1 (same  $\beta$  and anchor grids, five seeds, per-variant configuration selection) on the three settings of the mechanism analysis, and the reference models (Base, CEB, GPB, GPB-L) are re-trained in the same batch so every comparison is paired by seed.

**The ablation ladder.** *NCM-learn* removes the bottleneck entirely:  $z = \mu_\phi(x)$  is deterministic, prediction is the energy rule  $\text{logit}_k = -\|z - m_k\|^2/2$  on trainable prototypes  $m_k$ , and there is no posterior and no KL. *NCM-ortho* fixes the prototypes to the orthogonal frame  $a e_k$ : fixed geometry, still no IB. *CEB-energy* and *CEB-tied* mount the same energy readout (resp. the tied projection) on CEB’s trainable free reference: free geometry, free scale, restricted readout, with IB. *OPB-FS* and *EPB-FS* keep the orthogonal frame and the energy (resp. tied) readout but make the anchor radius learnable — per-class scales  $s_k$ , or a scalar  $\rho$  — initialized at the anchor scale, so only the fixed scale is switched off. Together with Base, CEB, GPB-L and GPB the ladder switches each factor independently: geometry (CEB-energy vs. OPB-FS), scale (OPB-FS vs. GPB), readout (GPB-L vs. GPB; CEB vs. CEB-energy), and the IB term itself (NCM-ortho vs. GPB).

| variant    | MNIST (Acc %)                            | ImageNet-100 (Acc %)                   | Housing ( $R^2$ )                             |
|------------|------------------------------------------|----------------------------------------|-----------------------------------------------|
| Base       | 98.3 $\pm$ 0.1                           | 83.1 $\pm$ 0.1                         | 0.762 $\pm$ 0.024                             |
| NCM-learn  | 98.2 $\pm$ 0.2                           | 82.2 $\pm$ 0.4                         | —                                             |
| NCM-ortho  | 98.1 $\pm$ 0.1 ( $a = 1$ )               | 82.3 $\pm$ 0.6 ( $a = 1$ )             | —                                             |
| CEB        | 98.4 $\pm$ 0.1 ( $\beta = 0.01$ )        | 82.8 $\pm$ 0.2 ( $\beta = 0.01$ )      | 0.789 $\pm$ 0.002 ( $\beta = 0.0001$ )        |
| CEB-energy | 98.2 $\pm$ 0.1 ( $\beta = 0.001$ )       | 81.6 $\pm$ 0.2 ( $\beta = 0.001$ )     | —                                             |
| GPB-L      | 98.4 $\pm$ 0.1 ( $\beta = 0.1, a = 12$ ) | 83.4 $\pm$ 0.7 ( $\beta = 1, a = 12$ ) | —                                             |
| GPB        | 98.4 $\pm$ 0.1 ( $\beta = 0.1, a = 12$ ) | 84.2 $\pm$ 0.1 ( $\beta = 1, a = 12$ ) | —                                             |
| OPB-FS     | 98.5 $\pm$ 0.0 ( $\beta = 0.1, a = 12$ ) | 84.3 $\pm$ 0.2 ( $\beta = 1, a = 12$ ) | —                                             |
| CEB-tied   | —                                        | —                                      | 0.781 $\pm$ 0.006 ( $\beta = 0.01$ )          |
| EPB-L      | —                                        | —                                      | 0.791 $\pm$ 0.003 ( $\beta = 0.0001, a = 6$ ) |
| EPB        | —                                        | —                                      | 0.788 $\pm$ 0.004 ( $\beta = 0.0001, a = 6$ ) |
| EPB-FS     | —                                        | —                                      | 0.787 $\pm$ 0.008 ( $\beta = 0.001, a = 6$ )  |

Table 3: Attribution ablation: best-configuration test metrics (mean  $\pm$  std over five seeds). Configurations are selected by the mean test metric, applied identically to every variant, so all comparisons remain paired by seed; the selection is slightly optimistic in level but shared across the table. One OPB-FS configuration (ImageNet-100,  $\beta = 10^{-4}$ ,  $a = 1$ ) diverged during training (free scales grow without penalty at negligible  $\beta$ ) and is excluded.

| factor                  | comparison            | MNIST                | ImageNet-100         | Housing              |
|-------------------------|-----------------------|----------------------|----------------------|----------------------|
| readout                 | GPB — GPB-L           | +0.02 [−0.05, 0.08]  | +0.82 [0.34, 1.17]   | —                    |
| readout (EPB)           | EPB — EPB-L           | —                    | —                    | −0.00 [−0.01, −0.00] |
| readout (free geometry) | CEB-energy — CEB      | −0.21 [−0.30, −0.12] | −1.19 [−1.50, −0.96] | —                    |
| geometry                | OPB-FS — CEB-energy   | +0.32 [0.26, 0.40]   | +2.67 [2.45, 2.92]   | —                    |
| geometry (EPB)          | EPB-FS — CEB-tied     | —                    | —                    | +0.01 [−0.00, 0.01]  |
| fixed scale             | GPB — OPB-FS          | −0.09 [−0.16, −0.02] | −0.04 [−0.07, −0.01] | —                    |
| fixed scale (EPB)       | EPB — EPB-FS          | —                    | —                    | +0.00 [−0.01, 0.01]  |
| IB increment*           | GPB — NCM-ortho       | +0.34 [0.22, 0.45]   | +1.89 [1.41, 2.36]   | —                    |
| prototype structure     | NCM-ortho — NCM-learn | −0.06 [−0.20, 0.11]  | +0.16 [−0.49, 0.82]  | —                    |
| prototype readout alone | NCM-learn — Base      | −0.16 [−0.28, −0.05] | −0.96 [−1.30, −0.69] | —                    |

Table 4: Factor-wise paired differences (accuracy points on MNIST and ImageNet-100,  $R^2$  on Housing; 95% bootstrap percentile CIs,  $B = 10,000$ , fixed seed, paired across the shared five seeds at each variant’s best configuration; positive values favor the first term). \*NCM-ortho fixes the frame at  $a e_k$  while GPB trains it, so the IB comparison additionally contains frame trainability.

**Where the gain comes from.** Table 3 reports the best configuration of each variant and Table 4 the factor-wise paired differences. On ImageNet-100 the decomposition is clean: the geometry contributes +2.67 points (OPB-FS — CEB-energy, CI [2.45, 2.92]), the IB term +1.89 (GPB — NCM-ortho, [1.41, 2.36]), and the restricted readout +0.82 (GPB — GPB-L, [0.34, 1.17]), while the fixed scale is neutral (−0.04, [−0.07, −0.01]). The readout effect flips sign on the free-geometry side, where the energy head costs CEB 1.19 points ([−1.50, −0.96]): the restricted readout pays only on the constructed geometry. Most informative is the negative result on the prototype readout alone: NCM-learn trails the plain backbone by 0.96 points ([−1.30, −0.69]), so the gain is not a nearest-class-mean effect in disguise — the prototypes need the IB objective and the orthogonal geometry to carry prediction. On MNIST the same ordering holds with small margins (geometry +0.32, IB +0.34, scale +0.09, readout  $\approx$  0), and on Cal. Housing every EPB variant ties within 0.01  $R^2$ , consistent with the flat EPB plateaus of Section 3.5. The conclusion is that the orthogonal geometry is the primary driver of the ImageNet-100 gain, and it acts *within* the IB framework: geometry, IB, and readout each contribute positively, in that order, and none of the bundled mechanisms is doing the work alone.

### 3.5 COMPRESSION-ACCURACY AND THE KL CERTIFICATE OVER $\beta$

The main table selects the best ( $\beta, a$ ) per method; the figures below report the compression-accuracy trade-off on one classification and one regression setting (ImageNet-100 and Cal. Housing, both with the MLP backbone). Each setting has two companion panels over the same  $\beta$  grid: the task metric against  $\beta$  (panels a–b), and the realized test-set  $\mathbb{E}[\text{KL}]$  — the certificate itself — against  $\beta$  (panels c–d). The certificate panels are what make a  $\beta$ -matched sweep interpretable: the two

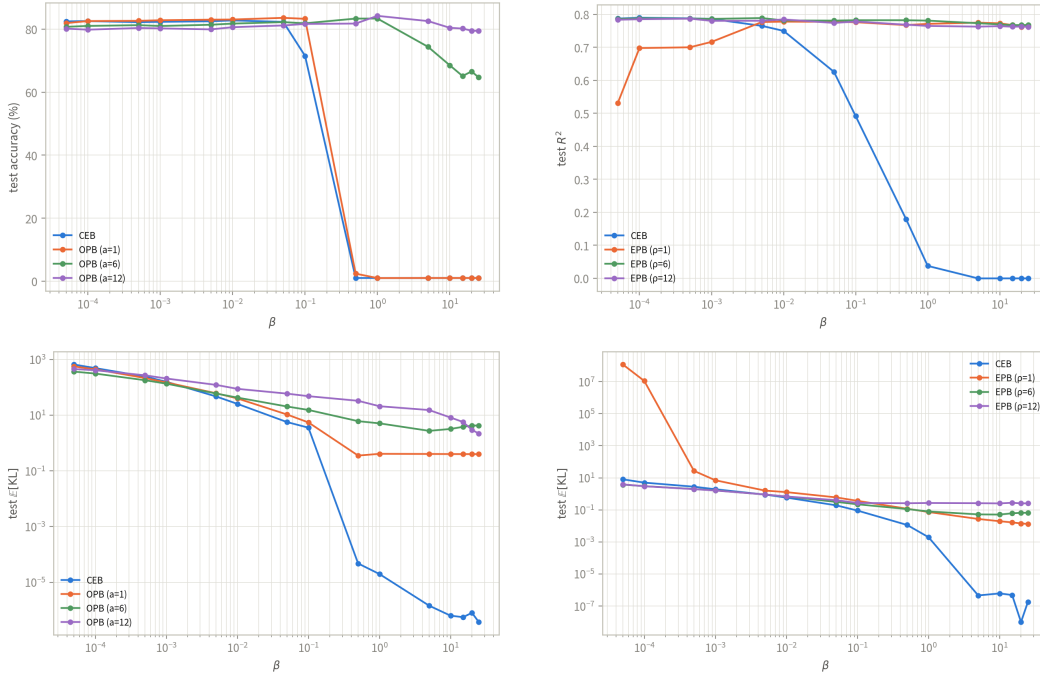


Figure 2: Compression-accuracy and the realized certificate over  $\beta$  (log scale),  $\beta \in \{5 \cdot 10^{-5}, \dots, 25\}$ ; CEB one curve per panel, GPB three ( $a$  on classification,  $\rho$  on regression, both  $\in \{1, 6, 12\}$ ). (a)–(b) Test accuracy (%) on ImageNet-100 (MLP) and test  $R^2$  on Cal. Housing (MLP) against  $\beta$ : OPB with  $a = 12$  holds the plateau out to  $\beta = 25$  while CEB and OPB ( $a = 1$ ) collapse. (c)–(d) The realized test-set  $\mathbb{E}[\text{KL}]$  against  $\beta$  (log scale) for the same runs: CEB’s certificate collapses with its reference, EPB ( $\rho = 1$ ) explodes near  $\beta = 5 \cdot 10^{-5}$ , and the larger anchors keep  $\mathbb{E}[\text{KL}]$  in a stable band.

methods reach different compressions at the same  $\beta$  — on Housing at  $\beta = 0.1$ , CEB sits at  $\mathbb{E}[\text{KL}] \approx 0.09$  while EPB ( $\rho = 12$ ) is at  $\approx 0.27$  — so the compression each point actually achieves is read off directly rather than assumed from  $\beta$ . No points are omitted: collapsed references and variance explosions are visible at the extremes of the sweep.

Figure 2 exhibits the two roles of the anchor scale across the  $\beta$  sweep, with panels (c)–(d) translating each point into its realized compression. (i) *Without a scale, the reference collapses.* On ImageNet-100, CEB’s accuracy erodes to 71.5% at  $\beta = 0.1$  ( $\mathbb{E}[\text{KL}] \approx 3.6$ ) and its reference escapes entirely at  $\beta \geq 0.5$  ( $\mathbb{E}[\text{KL}] \rightarrow 0$  with accuracy at chance, panel c); OPB at  $a = 1$  survives one grid step longer (83.3% at  $\beta = 0.1$ ) and collapses at the same end of the sweep. OPB ( $a = 6$ ) holds accuracy above 74% out to  $\beta = 5$  and falls to 64.7% at  $\beta = 25$ . On Housing, CEB drops to  $R^2 = 0.49$  at  $\beta = 0.1$  and below 0.1 by  $\beta \geq 0.5$ , while every EPB curve keeps  $R^2 \geq 0.76$  out to  $\beta = 25$ . Orthogonality alone therefore does not prevent the collapse: the frame is fixed up to rotation, but the anchor-energy classifier’s margin is  $a^2/\tau^2$  (Proposition 2.2), and at  $a = 1$  that margin is too small to carry a solution at strong compression. (ii) *The anchor scale controls where the collapse sets in, and at matched compression the geometry pays or ties.* On ImageNet-100, OPB ( $a = 12$ ) attains 84.2% at  $\beta = 1$  ( $\mathbb{E}[\text{KL}] \approx 21$ ), above CEB’s best point anywhere on its sweep (82.8% at  $\beta = 10^{-2}$ ), and still 79.4% at  $\beta = 25$  ( $\mathbb{E}[\text{KL}] \approx 2.2$ ); on Housing the four curves peak within half a point (CEB 0.789, EPB 0.786–0.789), but only EPB retains its peak into the strong-compression regime: the plateau is the flat EPB segment where  $R^2$  stays above 0.76 from  $\beta = 0.1$  out to  $\beta = 25$  while the certificate keeps shrinking.

A visible cost remains in the weak-compression regime: on ImageNet-100 at  $\beta = 10^{-4}$  ( $\mathbb{E}[\text{KL}] \gtrsim 300$ ), OPB trails CEB by one to three points ( $a = 6$ : 81.0 vs. 82.6;  $a = 12$ : 79.8 vs. 82.6): the price of the fixed-scale energy rule when compression is weak. On Housing the same regime shows EPB ( $\rho = 1$ ) dipping to  $R^2 = 0.53$  at  $\beta = 5 \cdot 10^{-5}$ , where its certificate explodes to  $\approx 10^8$  (panel d): with

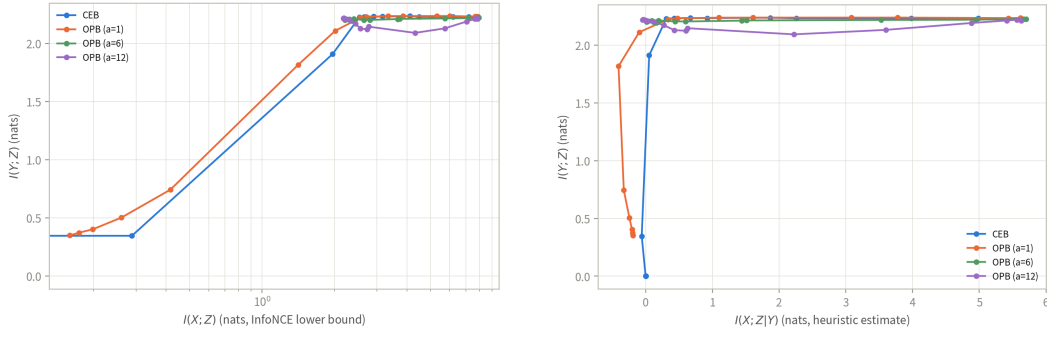


Figure 3: Information plane (a) and certificate plane (b) on MNIST over the  $\beta$  sweep, five runs per point, lines connect  $\beta$  in ascending order;  $x$  is log-scaled in (a), and the certificate axis in (b) switches to a linear scale whenever the estimated residual turns negative (OPB  $a = 1$  at  $\beta \geq 5$ , see text). CEB one curve per panel; OPB three ( $a \in \{1, 6, 12\}$ ). CEB follows the classic corner — compression and task information drop together past a critical  $\beta = 0.5$  with bimodal runs — and collapses to chance by  $\beta = 1$ ; OPB ( $a = 6, 12$ ) keeps accuracy flat out to  $\beta = 25$  with  $I(X; Z)$  saturating at  $\approx 2.2$ – $2.4$  nat, essentially all of it task-relevant (certificate  $\approx 0$ ). The Cal. Housing panels are deferred to Fig. 12 in the appendix.

a unit axis and negligible  $\beta$  the off-axis variance is unconstrained. The construction pays for itself in the strong-compression regime, where CEB is at chance.

These curves are empirical compression diagnostics, not a test of the zero-floor rate. For every benchmark distribution, Theorem 2.1 bounds anchor mismatch relative to any feasible comparator, but its ambiguity and approximation floor is unknown. Only in the deterministic exactly realizable special case does Corollary 2.1 reduce the bound to  $(\ln K + \epsilon)/\beta$ . At exact alignment, the Gaussian mixture’s accuracy is governed by the scale-to-noise ratio  $a/\tau$  (the nearest-anchor error probability is  $\Phi(-a/(\sqrt{2}\tau))$ ; appendix). For the benchmark runs, the realized KL and geometry are therefore inspected directly. With the classifier scale fixed at  $a/\tau^2$  by construction, the only remaining lever at full compression is  $a$  itself: the anchor scale converts the compression knob from a collapse dial into a plateau dial. The KL decomposition shows the two channels at work: OPB’s  $\mathbb{E}[\text{KL}]$  is dominated by the mean-mismatch term (anchor displacement;  $a = 12$ ,  $\beta = 1$ : 20.6 mean vs. 0.36 variance), while CEB’s reference absorbs compression by shrinking both terms toward zero: the reference chases the posterior, exactly the degeneracy the constructed geometry excludes (Lemma A.1). This decomposition motivates the direct mismatch diagnostics of Section 3.8.1; it does not by itself verify the comparator bound. The geometry transfer is characterized by the mismatch-dependent separation statement (Proposition 2.4) and the fixed-scale condition of Proposition A.4; consistent with that proposition’s scope, we do not claim that the map  $\beta \mapsto$  accuracy is monotone, only that the constructed scale keeps the plateau from collapsing.

### 3.6 INFORMATION-PLANE ANALYSIS

The certificate tells us *which* gap the KL measures; the panels below inspect *what* each method actually discards. On every sweep checkpoint we estimate the three quantities of the canonical information plane (Fig. 3):  $I(X; Z)$  by the InfoNCE lower bound (van den Oord et al., 2018) ( $\log B - \mathcal{L}_{\text{NCE}}$ ,  $B = 4096$ , one critic per run trained on the training split, the bound read off the test split with  $z \sim q(z|x)$ );  $I(Y; Z)$  by the classification lower bound  $H(Y) - \text{CE}(\text{logits})$ ; and the residual  $I(X; Z|Y)$  as the difference of the two — a heuristic, not a bound. The true residual is nonnegative by the data-processing inequality; negative plotted values mean the estimators, not the theory, gave way, and they occur exactly where the InfoNCE bound fails (see the caveats below). Consistent with Remark A.4, this section reports estimator-level observations on the sweep checkpoints and claims nothing about the exact plane.

(i) *The corner is sharp.* On MNIST, CEB moves from  $I(X; Z) = 2.53$  at 98.2% ( $\beta = 0.1$ ) to 1.96 at 86.0% ( $\beta = 0.5$ ) to 0.29 at 26.8% ( $\beta = 1$ ), and to chance beyond. The  $\beta = 0.5$  point is a critical point rather than a midpoint: the per-seed accuracy is bimodal ( $86.0 \pm 22.5\%$ ; one run

| Setting              | CEB             |                 | OPB (nearest $I(X; Z)$ ) |                 |                |
|----------------------|-----------------|-----------------|--------------------------|-----------------|----------------|
|                      | $I(X; Z)$       | Acc             | config                   | $I(X; Z)$       | Acc            |
| MNIST, $\beta = 0.1$ | $2.53 \pm 0.03$ | $98.2 \pm 0.1$  | $a = 1, \beta = 0.1$     | $2.55 \pm 0.01$ | $98.1 \pm 0.1$ |
| MNIST, $\beta = 0.5$ | $1.96 \pm 0.53$ | $86.0 \pm 22.5$ | $a = 1, \beta = 0.5$     | $2.01 \pm 0.03$ | $97.8 \pm 0.1$ |

Table 5: Matched-compression comparison on MNIST: each CEB checkpoint is paired with the OPB configuration of nearest estimated  $I(X; Z)$  (nats; mean  $\pm$  std over five runs). At matched compression the geometric prior retains strictly more of the task; the CEB  $\beta = 0.5$  entry is the bimodal critical point of the collapse transition (individual runs at 97% or 41%).

collapses to 41% while four sit at 96.5–97.8%). The two axes fall together — the reference absorbs the compression by absorbing the task information.

(ii) *The plateau has a floor at the task information.* OPB ( $a = 6, 12$ ) keeps MNIST accuracy at 98.1–98.6% for every  $\beta$  up to 25;  $I(X; Z)$  saturates at 2.2–2.4 nat while  $I(Y; Z)$  stays at  $\approx 2.2$  (against  $H(Y) = \ln 10 \approx 2.30$ ), and the certificate sits at  $\approx 0$  ( $a = 12, \beta = 15$ :  $0.00 \pm 0.21$ ;  $a = 6, \beta = 5$ :  $0.09$ ). The floor is the data-processing inequality made visible: at full compression the frame leaves room for exactly the task-relevant information and nothing else.

(iii) *At matched compression the geometry keeps more of the task.* Table 5 pairs each CEB point with the OPB configuration of nearest estimated  $I(X; Z)$ . At  $\approx 2.0$  nat, CEB ( $\beta = 0.5$ ) retains 86.0% while OPB ( $a = 1, \beta = 0.5$ ) retains 97.8%; at  $\approx 2.5$  nat ( $\beta = 0.1$ ) the two tie (98.2% vs. 98.1%) — the geometry pays nothing where compression is mild and carries the task where it is not.

*Estimator caveat.* One region must be read with care, and it is not used for any claim above: OPB ( $a = 1$ ) at  $\beta \geq 5$ . There the plotted estimates collapse ( $I(X; Z) \leq 0.42$  nat) while accuracy stays 97.6–97.8%; by Fano’s inequality the 2.3% error alone requires  $I(Y; Z) \geq H(Y) - h(0.023) - 0.023 \ln 9 \approx 2.1$  nat, so the leftward spur is a bound failure, not compression:  $z$  quantizes onto the ten anchors, near-duplicate positives saturate the InfoNCE objective (Poole et al., 2019), and overconfident errors inflate the cross-entropy.

### 3.7 ADVERSARIAL ROBUSTNESS

We evaluate robustness to white-box attacks following the CEB paper’s *targeted* variant (its Figure 4, bottom panels): targeted projected gradient descent (PGD, 20 steps, step size  $2.5\varepsilon$ /steps, random start) toward a single class, with success measured as the fraction of originally correct predictions flipped to that class. The target is digit 0, the most distinctive MNIST class (the analogue of Fashion-MNIST “trouser”), so that class confusion does not masquerade as robustness.  $\varepsilon$  is defined in the normalized pixel space (MNIST mean/std); we attack 1,000 test images and report the mean and standard deviation over runs. All stochastic models are attacked and evaluated on their *stochastic* path: each PGD step averages gradients over 10 samples of  $z \sim q_\phi(z|x)$  (expectation over transformation), and every prediction is the majority class under 10 MC samples. We use  $\varepsilon_\infty = 0.3$  and  $\varepsilon_2 = 6.0$  — the smallest  $L_2$  budget we tested at which the targeted attack engages the decision boundaries at all (at 2.0 it cannot reach the target region from any other class and sits at a near-zero floor for every model). Every curve is drawn only over the  $\beta$  values at which the model’s clean accuracy clears a shared 90% failure threshold under the same stochastic protocol: below it a configuration is counted as failed, and a failed model’s near-zero attack success is not robustness. The threshold binds CEB at  $\beta \leq 0.1$  (beyond, the reference collapses to chance) and OPB ( $a = 1$ ) at  $\beta \leq 1$  (beyond, its posterior retains substantial stochastic variance while its means pin to the radius-1 anchors, and the MC-sampled predictions fall to 0.59–0.84 clean accuracy even though the deterministic path keeps 0.98); OPB ( $a = 6, 12$ ) clears the threshold across the full grid.

Figure 4 shows the targeted-attack picture. Every curve falls as  $\beta$  grows, reproducing the CEB paper’s compression–robustness gradient under a protocol the deterministic path cannot game. Within CEB’s operating range the two geometries are comparable at matched  $\beta$  (at  $\beta = 0.1$  under  $L_\infty$ : 37.4% for both CEB and OPB,  $a = 12$ ), but the  $\beta$  axis again hides the compression gap — at  $\beta = 0.1$  CEB is the *more* compressed of the two (realized  $\mathbb{E}[\text{KL}] = 0.68$  vs. 1.23 for OPB,  $a = 12$ ). At compression-matched points the two geometries are at par within run-to-run variation: OPB ( $a = 12, \beta = 1$ ;  $\mathbb{E}[\text{KL}] = 0.55$ ) attains 30.6% vs. CEB’s 37.4% under  $L_\infty$ , a gap inside the seed spread. The decisive margin is the operating range. Under the shared 90% failure threshold, CEB’s



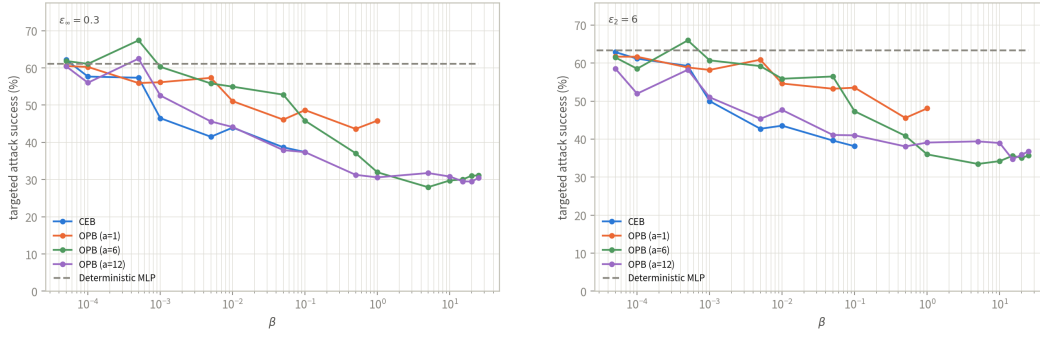


Figure 4: Targeted PGD attack success rate against  $\beta$  (log scale) on MNIST (MLP backbone): target class 0, 20 steps, EOT-10 gradients and 10-sample MC predictions on the stochastic path. (a)  $L_\infty$ ,  $\varepsilon = 0.3$ ; (b)  $L_2$ ,  $\varepsilon = 6.0$ . The dashed grey line is the deterministic MLP baseline. Every curve is drawn only over  $\beta$  values with clean accuracy above the shared 90% failure threshold (CEB:  $\beta \leq 0.1$ ; OPB  $a = 1$ :  $\beta \leq 1$ ;  $a = 6, 12$ : full grid). Every curve falls with  $\beta$ , and OPB keeps gaining robustness across its valid range: at  $\beta = 25$ , OPB ( $a = 6, 12$ ) reach 31.1%/30.5% — about 30 points below the baseline.

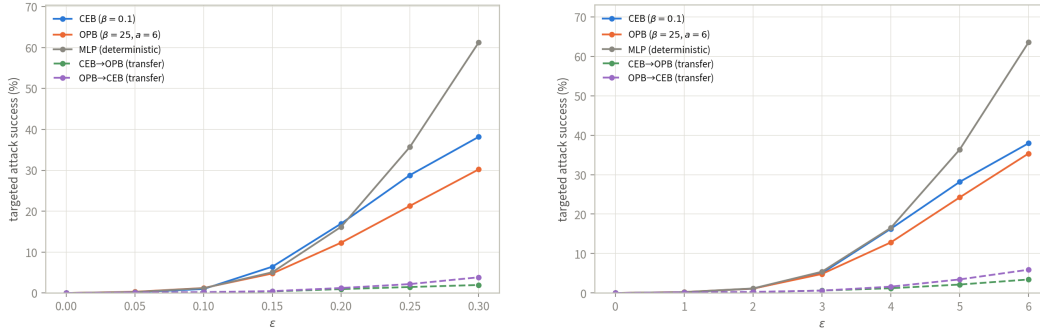


Figure 5: Transfer attacks (gradient-masking check) on MNIST (MLP backbone): targeted PGD adversaries generated on one model and evaluated on another, against  $\varepsilon$ . (a)  $L_\infty$ ; (b)  $L_2$ . Solid: white-box self-attacks of CEB ( $\beta = 0.1$ ), OPB ( $\beta = 25$ ,  $a = 6$ ) and the deterministic MLP baseline. Dashed: transfer attacks. CEB  $\rightarrow$  OPB stays an order of magnitude below OPB’s own white-box attack, so OPB’s extreme-compression robustness is not gradient masking; OPB  $\rightarrow$  CEB transfers little as well — adversarial directions found on the strongly compressed posterior are largely model-specific.

curve ends at  $\beta = 0.1$  and OPB ( $a = 1$ )’s at  $\beta = 1$ , while the larger anchors keep accuracy at  $\approx 0.98$  out to  $\beta = 25$ , where their targeted success has fallen to 31.1%/30.5% ( $a = 6/12$ ) under  $L_\infty$  and 35.7%/36.8% under  $L_2$  — about 30 points below the deterministic baseline and below anything CEB attains with its accuracy intact. The  $a = 1$  truncation is itself informative: at  $\beta \geq 5$  the small anchor’s posterior retains substantial stochastic variance while its means pin to the radius-1 anchors, so the stochastic path the robustness protocol evaluates on drowns in noise — the failure mode the larger anchors avoid. We do not over-claim: this is one dataset and one target class.

**Transfer attacks (gradient-masking check).** A white-box attack could fail simply because the target model’s gradients are degenerate, crediting dead models with robustness. We follow the CEB paper’s transfer protocol as the check: generate targeted PGD adversaries on one model (the same 20-step, EOT-10 recipe) and evaluate them on another (10-sample MC predictions), sweeping  $\varepsilon$  over 0.05–0.3 ( $L_\infty$ ) and 1–6 ( $L_2$ ). The models are CEB ( $\beta = 0.1$ , the last point of its operating range), OPB at maximum compression ( $\beta = 25$ ,  $a = 6$ ; realized  $\mathbb{E}[\text{KL}] = 0.40$ ), and the deterministic MLP baseline, whose white-box curve reaching 61% at  $\varepsilon_\infty = 0.3$  confirms that the attack itself is not broken.



Table 6: Mechanism validation: the constructed geometry pulls the posterior into the orthogonal frame (classification) and onto the isometric axis (regression). Geometric concentration: mean off-diagonal cosine of the posterior class-center Gram matrix (classification) / off-axis residual fraction (regression). Scale following: class-center radius  $\|\bar{c}\|$  against the anchor scale  $a$  (classification) / axial slope against  $\rho$  (regression). Cross-seed stability: std. over runs. Means  $\pm$  std. over five runs.

| Model (task)  | Task     | Geometric concentration | Scale following                 | Cross-seed stability |
|---------------|----------|-------------------------|---------------------------------|----------------------|
| CEB (MNIST)   | Classif. | 0.1205 $\pm$ 0.0026     | 5.14                            | 0.0201               |
| OPB (MNIST)   | Classif. | 0.0203 $\pm$ 0.0021     | 11.00 ( $a = 12$ )              | 0.0082               |
| CEB (Housing) | Regress. | 0.0233                  | 2.65 $\pm$ 0.59                 | 0.593                |
| EPB (Housing) | Regress. | 0.0002                  | 9.22 $\pm$ 0.15 ( $\rho = 12$ ) | 0.155                |

Figure 5 reads each transfer curve against its target’s own white-box attack, not against the source. CEB $\rightarrow$ OPB tops out at 2.0% ( $L_\infty$ ) and 3.4% ( $L_2$ ) — an order of magnitude below OPB’s own white-box attack (30.2%/35.4%) — so the extreme-compression robustness of OPB survives adversaries that never touch its gradients: it is geometry, not gradient masking. The reverse direction transfers little as well ( $\leq 3.8\%/5.9\%$ , against CEB’s own white-box 38.2%/37.9%); adversarial directions found on the strongly compressed posterior are largely model-specific, so neither model’s robustness is an artifact of the other’s gradients.

### 3.8 MECHANISM ANALYSIS: POSTERIOR GEOMETRY AND TIED PREDICTION

The main sweep compares predictive performance. This section separately evaluates the proposed mechanism on two representative settings, MNIST (MLP, classification) and California Housing (MLP, regression), trained at the matched compression  $\beta = 0.1$  with anchor scale  $a = \rho = 12$ . The table is a matched-configuration geometric snapshot; Section 3.8.1 then measures the theoretical mismatch directly across compression strengths.

**The prior.** CEB’s reference is arbitrary and drifts across seeds: the mean off-diagonal cosine of its learned class-prior table is  $0.119 \pm 0.002$ , and the scale of its learned regression axis varies by  $2.07 \pm 0.88$  across runs. The constructed prior has no such freedom: the OPB class priors are exactly orthogonal (off-diagonal cosine 0, pairwise distance  $\sqrt{2}a = 16.97$ ), and the EPB axis is fixed at  $\rho = 12.000$  with exact isometry through the origin, both verified at machine precision.

**The posterior.** Table 6 shows that this geometry transfers to the posterior. On classification the posterior class centers are nearly orthogonal (the mean off-diagonal Gram cosine is  $5.9\times$  lower than CEB’s) and sit at radius  $\|\bar{c}\| = 11.00$  (92% of  $a$ ), with  $2.5\times$  lower cross-seed drift; the Fisher ratio is comparable (15.3 vs. 15.7, reported as is). On regression the posterior lives essentially *on* the isometric axis: the off-axis residual fraction is  $141\times$  lower than CEB’s, and the axial slope tracks  $\rho$  with  $3.8\times$  lower cross-seed variation. The slope is 9.22 rather than  $\rho = 12$  because the encoder’s uncertainty shrinks the posterior along the axis (the slope equals  $\rho \cdot R_{\text{axis}}^2$ , as  $12 \times 0.78 \approx 9.4$ ), and the axis *direction* itself can rotate rigidly with the representation (mean direction cosine 0.03–0.06 for both models), so the discriminative stability evidence is the scale, not the direction. Figures 6 and 7 show both mechanisms directly.

**Prediction reads off the geometry.** Both heads of Section 2.2 have no free parameters. At the matched configuration the anchor-energy classifier attains 98.44% test accuracy against CEB’s 98.16% on MNIST, and the tied projection head attains  $R^2 = 0.778$  against CEB’s 0.491 on Housing: under this strong compression CEB collapses while the geometric readout retains the signal, consistent with the main-table GPB column (Section 3.3) and with the nearest-anchor analysis of the appendix. At this matched configuration, the evidence supports both links of the mechanism: the constructed prior is accompanied by the declared posterior geometry, and a zero-parameter rule that reads that geometry carries the prediction. This is mechanism evidence rather than a verification of the comparator inequality.

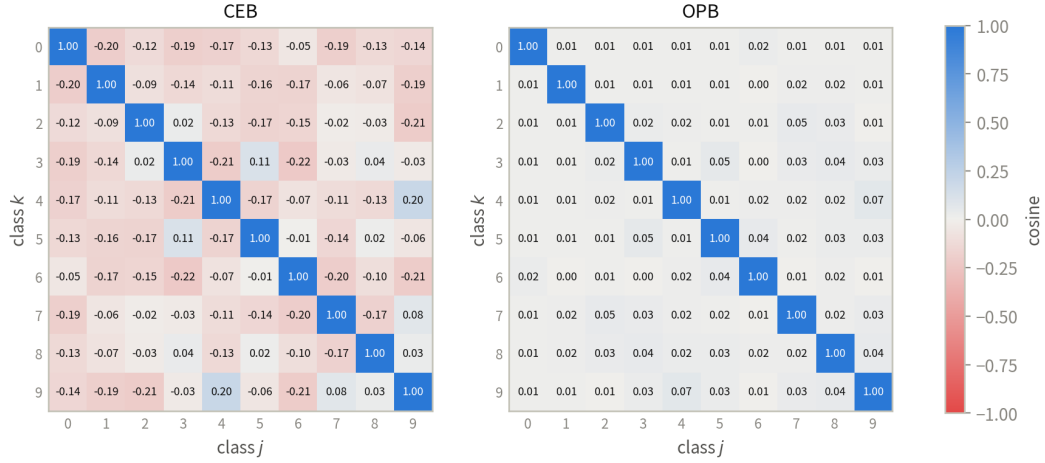


Figure 6: Pairwise-cosine Gram matrix of the posterior class centers on MNIST, averaged over five runs. OPB is nearly diagonal: the posterior centers are orthogonal; CEB shows no such structure.

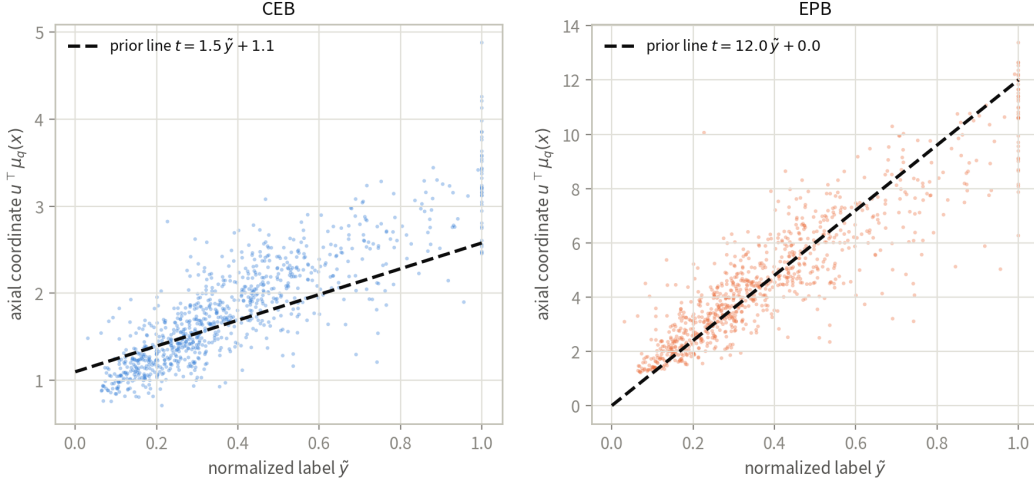


Figure 7: The axial coordinate  $u^\top \mu_q(x)$  against the normalized label on Housing (one representative run per model). The EPB posterior hugs the prior line  $\rho \tilde{y}$ , shrunk toward  $\rho \mathbb{E}[\tilde{y} | x]$  by the encoder’s uncertainty.

### 3.8.1 DIRECT MISMATCH ANALYSIS ACROSS COMPRESSION STRENGTHS

The preceding snapshot uses geometric proxies at one operating point. We now measure the mismatch in Proposition 2.4 directly over the complete compression grids of ImageNet-100 (MLP) and Cal. Housing (MLP), with five seeds per point and no missing checkpoints. For each checkpoint, we set  $\hat{\tau}_{\max}^2$  to the largest conditional-reference variance realized over the class table (classification) or test labels (regression), and compute

$$\hat{D} = \frac{1}{N} \sum_{n=1}^N \frac{\|\mu_\phi(x_n) - m_\theta(y_n)\|^2}{2\hat{\tau}_{\max}^2}. \quad (36)$$

This is a finite-test diagnostic using a checkpoint-wise empirical variance bound, not an estimate of a population-wide constant. KL uses each sample’s actual diagonal prior variance and is reported separately. CEB is included only as a descriptive mismatch reference; because its class means have no declared separation, no GPB separation bound is applied to it. Full definitions and pair-level diagnostics are in Appendix A.6.

**Classification.** On ImageNet-100, OPB’s mismatch decreases monotonically with compression at fixed scale: for  $a = 6$ ,  $\hat{D}$  falls from 62.3 at  $\beta = 5 \cdot 10^{-5}$  to 2.54 at  $\beta = 25$ . For each class pair  $(i, j)$  we compare the posterior-center distance  $S_{ij} = \|m_i - m_j\|$  with the raw empirical lower bound

$$\widehat{\text{LB}}_{ij} = \sqrt{2}a - \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_i} - \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_j}. \quad (37)$$

At  $\beta = 25$ , this bound is positive for 96.9% of class pairs at  $a = 12$  (mean lower bound 3.95), 62.3% at  $a = 6$ , and none at  $a = 1$ . The last case is informative rather than contradictory: with  $a = 1$  the posterior centers collapse at  $\beta = 25$  (mean  $S_{ij} = 0$ ), so the small declared margin is entirely consumed by mismatch. At the other end,  $a = 6$  overshoots the frame at  $\beta = 5 \cdot 10^{-5}$  ( $S/(\sqrt{2}a) = 1.39$ ); this ratio falls to 0.51 at  $\beta = 25$  as compression pulls the posterior inward. Every applicable seed-checkpoint pair has non-negative separation slack to numerical tolerance, and the Jensen check on every class also holds. These are consistency checks of an unconditional inequality, not independent evidence for an optimization rate.

**Regression.** On Housing with  $\rho = 6$ ,  $\hat{D}$  falls from 0.264 at  $\beta = 5 \cdot 10^{-5}$  to 0.026 at  $\beta = 25$ , while  $R^2$  changes only from 0.786 to 0.767. This separates the mechanism measurement from the performance comparison: a tenfold mismatch reduction need not imply a comparable task-metric gain. The mismatch is almost entirely axial (a representative axial contribution is about 0.2, against an off-axis contribution of about 0.004), and the normalized axial slope remains 0.73–0.77: the posterior follows the declared axis but at roughly three quarters of its scale. With finite label bins and the quantization correction of Remark A.2, 43% of bin pairs retain a positive lower bound. The mean bound is negative and the actual bin-center distances lie about 25% below the prior isometry, so the bound is consistent but conservative. Small- $\beta$ , large- $\rho$  variance-explosion checkpoints are retained; their enlarged  $\hat{\tau}_{\max}^2$  can reduce the normalized  $\hat{D}$  even while the KL variance term grows, which is why the components in Fig. 8 must be read together.

The two tasks expose complementary limits of the geometric claim. Classification has a discrete declared margin, and sufficiently large anchors leave an informative positive separation certificate for most class pairs. Regression transfers direction more clearly than scale, and its finite-bin lower bound is often vacuous. Neither observation turns benchmark accuracy into a test of Theorem 2.1; they directly evaluate the realized mismatch on which the unconditional Proposition 2.4 depends.

## 4 RELATED WORK

**Information bottleneck and conditional compression.** The information bottleneck (IB) principle formulates representation learning as retaining information about a target while discarding information about the input (Tishby et al., 2000). Deep VIB turns this principle into a neural objective by using a reparameterized stochastic encoder and a variational marginal prior (Alemi et al., 2017). Nonlinear and non-parametric bottleneck variants study alternative variational bounds and distance-based estimates of the information in a representation (Kolchinsky et al., 2019). The deterministic-scenario analysis of Kolchinsky & Tracey (2019) further shows that the usual IB multiplier can fail to sweep an interior information-accuracy trade-off when labels are deterministic; the squared-compression baseline in our experiments follows this line of work.

The conditional entropy bottleneck (CEB) replaces the marginal reference of VIB by a label-conditional reference and thereby targets conditional residual information (Fischer, 2020). Comparisons of variational bounds clarify when this conditional reference is tighter than a marginal one (Geiger & Fischer, 2020). CEB has also been applied to large-scale visual representations and robustness evaluations (Fischer & Alemi, 2020; Lee et al., 2021), while multivariate extensions retain the same variational bottleneck viewpoint (Abdelaleem et al., 2025). These methods leave the relationships among distinct conditional reference means unspecified. GPB keeps the CEB posterior and its single conditional KL, but restricts the reference family itself: OPB declares a class frame and EPB declares a label-distance-preserving axis.

**Information bottleneck for robust representations.** Several works use a bottleneck to suppress nuisance or non-robust information. The HSIC bottleneck adds layer-wise dependence penalties and analyzes adversarial robustness (Wang et al., 2021). Adversarial Information Bottleneck explicitly

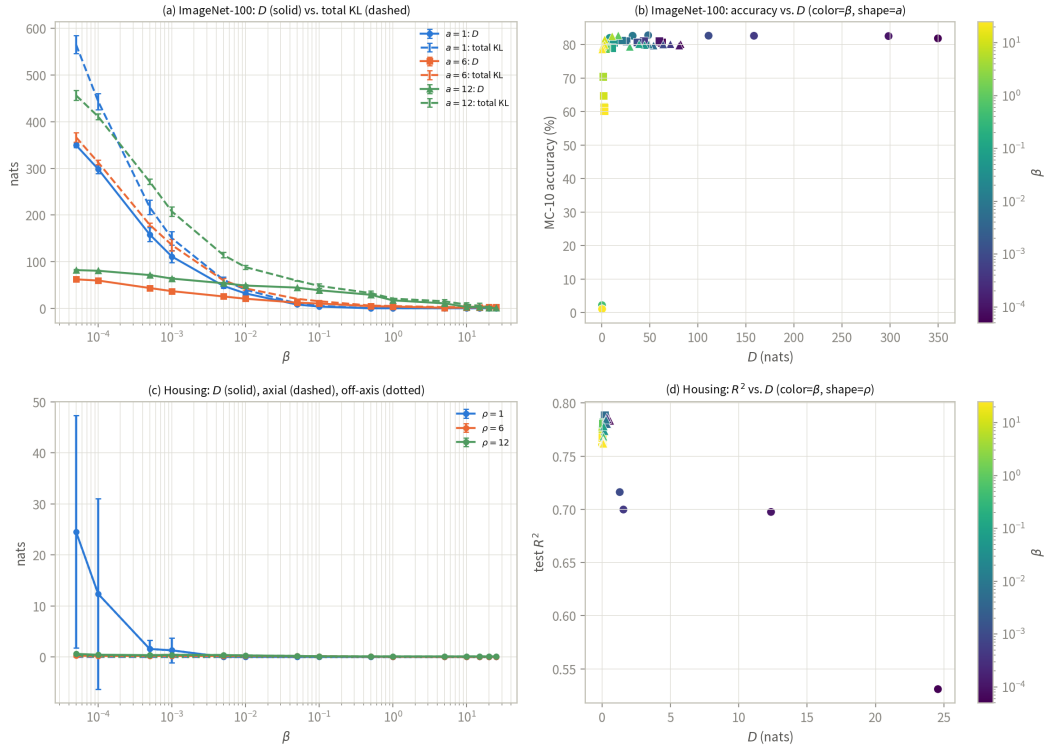


Figure 8: Direct mismatch diagnostics over the full compression sweeps. (a) ImageNet-100 anchor mismatch  $\hat{D}$  and total KL against  $\beta$ , by anchor scale. (b) MC-10 accuracy against  $\hat{D}$ ; color denotes  $\beta$  and marker shape the anchor scale. (c) Cal. Housing mismatch against  $\beta$ , decomposed into axial and off-axis contributions. (d) Test  $R^2$  against  $\hat{D}$ . Means and standard deviations are over five seeds; no checkpoint is omitted.

incorporates adversarial objectives (Zhai & Zhang, 2022), and an NLP-specific variant separates task-relevant robust features from manipulable features (Zhang et al., 2022). Causal Information Bottleneck uses a causal decomposition for the same robust-feature goal (Hua et al., 2022). These approaches modify the information criterion, feature selection, or training examples. In contrast, the geometric constraint in GPB is placed on the conditional prior and is evaluated with the ordinary CEB KL; the robustness experiments therefore test whether a declared prior geometry is transferred to the posterior rather than introducing adversarial training as an additional objective.

**Geometric classification representations.** Neural-collapse studies document an approximately simplex-ETF arrangement of class means and classifier directions near the terminal phase of training (Papayan et al., 2020; Zhu et al., 2021). The connection between neural collapse, supervised contrastive learning, and an information bottleneck has also been analyzed explicitly (Wang & Palmer, 2023). Other works impose related geometry directly: orthogonal classifier weights can improve adversarial robustness (Xu et al., 2022); ETF classifiers have been used for early exit (Ji et al., 2023); and fixed or hierarchy-aware frames have been used to reduce classifier bias or mistake severity (Tong & Davis, 2023). Neural Collapse by Design trains features and prototypes jointly on the unit hypersphere so that the global optimum of the NONL contrastive objective is exactly the simplex ETF (Koromilas et al., 2026); we include it as a classification-only comparison baseline. Equidistant prototype embeddings provide an earlier prototype-based classification example (Deng et al., 2014).

These works constrain features, classifier weights, or prototypes through a classification loss or a fixed readout. OPB addresses a different object: the label-conditional Gaussian prior in the bottleneck. Its polar/QR frame is recomputed from the prior encoder, and the anchor-energy classifier reads the same frame without introducing an independent classifier scale. Thus the geometric class code and the conditional KL are coupled during training.

**Distance-preserving regression and geometric priors.** Isometric representation learning has traditionally focused on preserving distances on a data manifold, including deep extensions of isomap-style mappings (Pai et al., 2022). Orthogonal label regression has also been used as a discriminative device in domain adaptation (Luo et al., 2023). These methods motivate using a declared label metric, but they do not formulate that metric as the mean map of a conditional Gaussian information-bottleneck prior. EPB uses the scalar label case: after standardization, prior means lie on a normalized axis with a fixed scale, so latent prior distances equal the declared label distances up to that scale. The regression information-bottleneck literature, including divergence-based bounds, provides complementary compression objectives rather than this geometric prior construction (Yu et al., 2024). AdaCap pairs a permutation-based contrastive loss with a Tikhonov closed-form output layer for small-sample tabular regression (Belucci et al., 2026); we include it as a regression-only comparison baseline.

**Geometry-aware information bottlenecks.** Recent work has used the phrase geometric information bottleneck in several distinct settings. A variational recommender learns a geometric prior for explanation generation (Yan et al., 2024); a data-scarcity formulation penalizes latent curvature and intrinsic dimension (Katende, 2025); and activation steering uses orthogonal subspaces to control language-model interventions (Doan et al., 2026). A concurrent analysis studies the relationship between information compression and class-wise geometric compression in CEB networks (Adilova et al., 2026). These studies make geometry an important representation concern, but they do not use the same conditional Gaussian prior for both compression and prediction, nor do they provide the OPB/EPB alignment-to-separation construction considered here.

## AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for drafting and editing portions of the code and of this manuscript. All AI-assisted content was manually reviewed and verified by the authors; experimental results and theoretical claims were produced and checked by our own code and analysis. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

## REPRODUCIBILITY STATEMENT

The code, hyperparameter grids, and evaluation protocol are described in Section 3.1; the geometric prior is stateless (no EMA or prototypes) and the orthogonalization is exact wherever defined (Assumption 2.1). All datasets are public, with preprocessing steps given in the release. Proofs of all statements are collected in the appendix. Statistical claims use the paired bootstrap confidence intervals and non-inferiority margin described in Appendix A.8.

## ACKNOWLEDGMENTS

## REFERENCES

- Eslam Abdelaleem, Ibrahima Ndiour, and Abhishek Bhattacharyya. Deep variational multivariate information bottleneck. *Journal of Machine Learning Research (JMLR)*, 2025.
- Linara Adilova, Henning Petzka, Asja Fischer, and Bernhard C. Geiger. Geometric and information compression of representations in deep learning, 2026.
- Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. In *International Conference on Learning Representations (ICLR)*, 2017.
- Bruno Belucci, Karim Lounici, and Katia Meziani. Adacap: An adaptive contrastive approach for small-data neural networks. In *Proceedings of the European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, 2026.

- Weihong Deng, Jiani Hu, Xiuzhuang Zhou, and Jun Guo. Equidistant prototypes embedding for single sample based face recognition with generic learning and incremental learning. *Pattern Recognition*, 2014. doi: 10.1016/j.patcog.2014.06.020.
- Toan Doan, Thin Nguyen, and Sunil Gupta. A geometric information bottleneck for activation steering, 2026.
- Ian Fischer. The conditional entropy bottleneck. *Entropy*, 22(9):999, 2020.
- Ian Fischer and Alexander A. Alemi. CEB improves model robustness. *Entropy*, 22(10):1081, 2020. doi: 10.3390/e22101081.
- Bernhard C. Geiger and Ian Fischer. A comparison of variational bounds for the information bottleneck functional. *Entropy*, 22(11):1229, 2020. doi: 10.3390/e22111229.
- Huan Hua, Jun Yan, Xi Fang, Weiquan Huang, Huilin Yin, and Wan-cheng Ge. Causal information bottleneck boosts adversarial robustness of deep neural network, 2022.
- Yixin Ji, Jikai Wang, Juntao Li, Qiang Chen, Wenliang Chen, and Min Zhang. Early exit with disentangled representation and equiangular tight frame. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. doi: 10.18653/v1/2023.findings-acl.889.
- Ronald Katende. Geometry as a missing axis of representation quality: The variational geometric information bottleneck under data scarcity, 2025.
- Artemy Kolchinsky and Brendan D. Tracey. Caveats for information bottleneck in deterministic scenarios. In *International Conference on Learning Representations (ICLR)*, 2019.
- Artemy Kolchinsky, Brendan D. Tracey, and David H. Wolpert. Nonlinear information bottleneck. *Entropy*, 21(12):1181, 2019.
- Panagiotis Koromilas, Theodoros Gianakopoulos, Mihalis A. Nicolaou, and Yannis Panagakis. Neural collapse by design: Learning class prototypes on the hypersphere, 2026.
- Kuang-Huei Lee, Anurag Arnab, Sergio Guadarrama, John Canny, and Ian Fischer. Compressive visual representations, 2021.
- Lingkun Luo, Shiqiang Hu, and Liming Chen. Discriminative noise robust sparse orthogonal label regression-based domain adaptation. *International Journal of Computer Vision*, 2023. doi: 10.1007/s11263-023-01865-z.
- Gautam Pai, Alex Bronstein, Ronen Talmon, and Ron Kimmel. Deep isometric maps. *Image and Vision Computing*, 2022. doi: 10.1016/j.imavis.2022.104461.
- Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences (PNAS)*, 117(40):24652–24663, 2020.
- Ben Poole, Sherjil Ozair, Aaron van den Oord, Alexander A. Alemi, and George Tucker. On variational bounds of mutual information. In *International Conference on Machine Learning (ICML)*, 2019.
- Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Proc. 37th Allerton Conference on Communication, Control, and Computing*, 2000.
- Liang Tong and Jim Davis. Inducing neural collapse to a fixed hierarchy-aware frame for reducing mistake severity. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023. doi: 10.1109/ICCV51070.2023.00139.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. 2018.
- Siwei Wang and Stephanie E. Palmer. Towards understanding neural collapse in supervised contrastive learning with the information bottleneck method, 2023.

- Zifeng Wang, Tong Jian, Aria Masoomi, Stratis Ioannidis, and Jennifer Dy. Revisiting hilbert-schmidt information bottleneck for adversarial robustness, 2021.
- Cong Xu, Xiang Li, and Min Yang. An orthogonal classifier for improving the adversarial robustness of neural networks. *Information Sciences*, 2022. doi: 10.1016/j.ins.2022.01.039.
- Hanqi Yan, Lin Gui, Menghan Wang, Kun Zhang, and Yulan He. Explainable recommender with geometric information bottleneck. *IEEE Transactions on Knowledge and Data Engineering*, 2024. doi: 10.1109/TKDE.2024.3350447.
- Shujian Yu, Xi Yu, Sigurd Lokse, Robert Jenssen, and Jose C. Principe. Cauchy-schwarz divergence information bottleneck for regression, 2024.
- Penglong Zhai and Shihua Zhang. Adversarial information bottleneck. *IEEE Transactions on Neural Networks and Learning Systems*, 2022. doi: 10.1109/TNNLS.2022.3172986.
- Cenyuan Zhang, Xiang Sean Zhou, Yixin Wan, Xiaoqing Zheng, Kai-Wei Chang, and Cho-Jui Hsieh. Improving the adversarial robustness of NLP models by information bottleneck. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022. doi: 10.18653/v1/2022.findings-acl.284.
- Zhihui Zhu, Tianyu Ding, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, and Qing Qu. A geometric analysis of neural collapse with unconstrained features, 2021.

## A DEFERRED PROOFS, CONSTRUCTION DETAILS, AND SCOPE

### A.1 FAILURE DIAGNOSIS

**Proposition A.1** (Dead knob, following Caveat 1 of Kolchinsky & Tracey (2019)). *Assume  $Y = f(X)$ . For every  $\gamma > 0$ , the CEB objective  $L_\gamma(Z) := I(X; Z|Y) - \gamma I(Y; Z)$  is minimized exactly on the corner family  $\{Z : I(Y; Z) = H(Y), I(X; Z|Y) = 0\}$ , and the map  $\gamma \mapsto (I(X; Z^*|Y), I(Y; Z^*))$  is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

*Proof sketch (see Kolchinsky & Tracey (2019) for the full argument).* By the chain rule,  $L_\gamma(Z) = I(X; Z) - (1 + \gamma)I(Y; Z)$ ; the data-processing inequality gives  $L_\gamma \geq -\gamma H(Y)$ , attained exactly at the corner  $Z = Y$ . In the Caveats paper’s maximizing convention this is their Eq. 7 with  $\beta' = 1/(1 + \gamma) \in (0, 1)$ , the range in which every maximizer is pinned at the copy corner.  $\square$

The corner is a single point of the information plane and an entire equivalence class of representations; the objective is exactly indifferent within the class, and neither the objective nor the certificate records which member the optimizer reaches. In the deterministic scenario this corner is the MNI point, so CEB’s declared target and the degenerate attractor coincide. The Caveats paper repairs this by squaring the compression term; our anchored reference supplies the same strict convexity by a different mechanism, without squaring: a quadratic displacement term built into the KL by construction (Proposition A.4 below). We present this as a mechanism connection to the Caveats fix, not as a new solution of the deterministic degeneration.

*Proof of Proposition 2.1.* For any matched pair ( $b = q(z|y)$ ), the gap identity of Eq. equation 1 gives Eq. equation 2; the matching condition is stated per class and no term of the certificate couples two distinct class priors, so the certificate value carries no information about the pairwise geometry of  $\{q(z|y)\}$ . Note that  $I(X; Z|Y)$  is generally non-zero; it vanishes only under the conditional independence  $Z \perp X | Y$ , a property of the encoder, not of the prior.  $\square$

**Proposition A.2** (Retention blindness on  $T_\alpha$ ). *On the manifold  $T_\alpha = B_\alpha \cdot Y$  of Kolchinsky & Tracey (2019) (their Eq. 6), a family that depends on  $Y$  alone,  $I(X; T_\alpha|Y) = 0$  for every  $\alpha$ , and with the optimal backward encoder and classifier,  $\text{ReX}(T_\alpha) = 0$  and  $\text{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha$ : the certificate reads 0 from the total collapse ( $\alpha = 0$ ) to the full copy ( $\alpha = 1$ ), while the retained label information  $I(Y; T_\alpha)$  sweeps the whole axis. Tightness of the certificate is therefore*

compatible with retaining all, part, or none of  $Y$ : the certificate records nothing about how much label information the code retains.

*Proof.*  $T_\alpha$  is a function of  $Y$  alone, hence  $T_\alpha \perp X \mid Y$  and  $I(X; T_\alpha \mid Y) = 0$ . By Proposition 2.1 with the true backward encoder the certificate is tight,  $\text{ReX} = I(X; T_\alpha \mid Y) = 0$ , and the optimal classifier attains  $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$ , i.e.  $\text{CE} = H(Y|T_\alpha) = \delta_\alpha$ .  $\square$

*Remark A.1* (The objective on  $T_\alpha$  is  $\beta$ -independent and strictly prefers the full copy). On  $T_\alpha$  the compression term contributes nothing ( $\text{ReX} \equiv 0$ ), so the objective reads  $L = \text{CE} + \beta \text{ReX} = \delta_\alpha$ , independent of  $\beta$  and strictly minimized at  $\alpha = 1$  (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits.

*Scope of the comparison argument.* Theorem 2.1 applies to arbitrary  $P_{XY}$  and every feasible comparator in the fitted model class; it does not require  $q(z|x) = p_\theta(z|y)$ . Its usefulness depends on the comparator KL. Label ambiguity, limited encoder capacity, and covariance misspecification produce the non-zero floor made explicit in Proposition 2.5. The stronger statement that this floor is zero still requires the assumptions of Corollary 2.1: deterministic labels, exact posterior–prior matching, and a compatible head. We do not assume those exact-realizability premises for the real-data benchmarks. Their performance comparison evaluates utility, while their mechanism analysis evaluates realized mismatch and geometry. The controlled known-noise experiment instead evaluates the ambiguity term where its population value is analytic, including the deterministic distribution as its zero-floor special case.

## A.2 PROOFS OF THE ALIGNMENT GUARANTEES

*Proof of Proposition 2.3.* For fixed  $\theta$ , apply the gap identity of Eq. equation 1 to the conditional distribution  $p_\theta(z|y)$  and the posterior  $q_\phi$ .  $\square$

*Proof of Proposition 2.4.* For (a), Jensen gives  $\|m_k - m_\theta(k)\| \leq \sqrt{2\tau_{\max}^2 D_k}$ , and the triangle inequality on  $m_i - m_j = (m_i - m_\theta(i)) + (m_\theta(i) - m_\theta(j)) + (m_\theta(j) - m_j)$  gives Eq. equation 16. For (b), the same two inequalities hold pointwise for  $P_Y$ -a.e.  $y$ : Jensen on the regular conditional expectation gives  $\|m(y) - m_\theta(y)\| \leq \sqrt{2\tau_{\max}^2 D(y)}$  (the norm is convex), and the triangle inequality gives Eq. equation 18; intersecting the two measure-one sets of valid  $y$ ’s yields the almost-everywhere pair statement.  $\square$

*Remark A.2* (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of part (b) are estimated by binning the label axis into intervals  $B_i$  with representative labels  $\bar{y}_i$  and centers  $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$ ; the bin diameter adds a quantization error bounded by  $\rho_{\max} \text{diam}(B_i) + \rho_{\max} \text{diam}(B_j)$  (by the upper bound of Eq. equation 5), which vanishes as the bins shrink. The separation statement itself is pointwise and needs no binning.

*Proof of Theorem 2.1.* Since  $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$ , the Gaussian KL contains a mean term at least  $\|\mu_\phi(x) - m_\theta(y)\|^2 / (2\tau_{\max}^2)$ . Hence  $\mathcal{D}(\hat{w}) \leq \mathcal{K}(\hat{w})$ . For any feasible  $w_0$ , near-optimality and non-negativity of the prediction loss give

$$\beta \mathcal{K}(\hat{w}) \leq \mathcal{R}(\hat{w}) + \beta \mathcal{K}(\hat{w}) \leq \mathcal{R}(w_0) + \beta \mathcal{K}(w_0) + \varepsilon.$$

Division by  $\beta$  proves Eq. equation 21; substituting  $\mathcal{K}(w_0) \leq \delta_{\text{app}}$  and  $\mathcal{R}(w_0) \leq C_{\text{app}}$  proves Eq. equation 22.  $\square$

*Proof of Proposition 2.5.* For OPB let  $A_Y := a q_{0,Y}$ . Conditional on  $X$ , orthonormality gives  $\mathbb{E}[A_Y \mid X] = a \sum_k \eta_k(X) q_{0,k}$  and

$$\mathbb{E}[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X] = a^2 (1 - \|\eta(X)\|_2^2).$$

The conditional bias–variance identity then yields

$$\mathbb{E}\|h(X) - A_Y\|^2 = a^2 \mathbb{E}[1 - \|\eta(X)\|_2^2] + \mathbb{E}\left\|h(X) - a \sum_k \eta_k(X) q_{0,k}\right\|^2.$$



Adding the diagonal-Gaussian covariance term  $V_0$  proves Eq. equation 24. For EPB set  $A_Y := \rho \tilde{Y} u_0$ . Then  $\mathbb{E}[A_Y | X] = \rho \nu(X) u_0$  and  $\mathbb{E}[\|A_Y - \mathbb{E}[A_Y | X]\|^2 | X] = \rho^2 \text{Var}(\tilde{Y} | X)$ . The same identity plus  $V_0$  proves Eq. equation 25.  $\square$

*Proof of Corollary 2.1.* For classification, the assumed comparator  $q(z|x) = p_\theta(z|y)$  has  $\mathcal{K}(w_0) = 0$ . Its Bayes cross-entropy is  $C_G = H(Y | Z_{\text{align}}) \leq \ln K$ ; with shared covariance the determinant factors cancel and its posterior is the anchor-energy rule. Applying Eq. equation 21 gives Eq. equation 27. For regression the aligned comparator again has zero KL. Under the tied head its error is  $(\tau/\rho)\eta$ ,  $\eta \sim \mathcal{N}(0, 1)$ , so its standardized squared risk is  $\tau^2/\rho^2$ ; the same theorem gives Eq. equation 28.  $\square$

*Proof of Corollary 2.2.* Theorem 2.1 gives  $\mathbb{E}[D(Y)] \leq B_0$ . For (a),  $D_k \leq B_0/\pi_k$ ; substituting this into Proposition 2.4(a) gives Eq. equation 29. For (b), Markov’s inequality gives  $P_Y(D(Y) > t) \leq B_0/t$ ; on  $S_t$  the pointwise transfer of Proposition 2.4(b) applies with  $D(y), D(y') \leq t$ .  $\square$

*Proof of Proposition 2.6.* By Jensen’s inequality the left side is at least  $\frac{1}{2\tau^2} \sum_k \pi_k \|c - a q_{\theta,k}\|^2 = \frac{1}{2\tau^2} (\|c\|^2 - 2a c^\top \sum_k \pi_k q_{\theta,k} + a^2)$ . Minimizing over  $c$  gives  $c = a \sum_k \pi_k q_{\theta,k}$ ; by orthonormality  $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$ , yielding Eq. equation 31.  $\square$

*Remark A.3* (What the framework constrains (and what it does not)). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame  $\{q'_k\}$  and the class-constant posterior  $q(z|Y = k) = \mathcal{N}(a q'_k, \tau^2 I_d)$ , the trainable frame can rotate to match ( $q_{\theta,k} \rightarrow q'_k$ ), yielding  $\text{KL} = 0$ ,  $I(X; Z|Y) = 0$ , and  $I(Y; Z) > 0$ : a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. In the fixed-covariance canonical instance, the remaining freedom at certificate zero is the frame rotation. The covariance-relaxed real-data protocol has an additional but log-clamped matching channel; the comparison theorem applies with the resulting  $\tau_{\max}^2$ , and Section 3.8.1 reports that channel rather than treating it as fixed.

### A.3 CONSTRUCTION DETAILS

**Polar permutation equivariance.** For any label permutation matrix  $P$ ,

$$Q_\theta(U_\theta P) = U_\theta P (P^\top U_\theta^\top U_\theta P)^{-1/2} = U_\theta (U_\theta^\top U_\theta)^{-1/2} P = Q_\theta(U_\theta) P, \quad (38)$$

since  $PP^\top = I_K$  and the inverse square root conjugates under  $P$ . Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role.

**Proposition A.3** (Orthogonal class-prior geometry by construction). *Under Eqs. equation 8–equation 9, the class means  $a_k = a q_{\theta,k}$  satisfy Eq. equation 10; equivalently, the class-mean Gram matrix is  $a^2 I_K$ : under Assumption 2.1, the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.*

*Proof.* Eq. equation 10 follows from  $Q_\theta^\top Q_\theta = I_K$  after scaling by  $a$ ; in particular  $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$ .  $\square$

**Initialization, monitoring, and re-initialization.** The posterior logvar head is zero-initialized ( $\sigma^2 = \tau^2 = 1$  at initialization, so training starts at  $\text{KL} \approx \frac{1}{2\tau^2} a^2$ ); the prior encoder is identity-initialized, so the initial raw matrix is  $U_\theta = [e_1, \dots, e_K]$ , full column rank, with a stable polar-factor backward (the verified condition of Assumption 2.1). The full-rank condition is *verified, not imposed*:  $\sigma_{\min}(U_\theta)$  is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective, which remains Eq. equation 11 exactly. Near rank deficiency the polar factor’s gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

**EPB axis choices.** Three variants of the direction: (A) *fixed axis*,  $u = [1, 0, \dots, 0]$ , with no trainable prior direction: the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*,  $u = W/\|W\|$  with  $W$  trained and normalized at every step, our main variant, which keeps the scale  $\rho$  exact while allowing the model to choose the axis; (C) *unconstrained linear axis*,  $\mu_p(y) = W\tilde{y}$ : the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and  $\tilde{y}$  is already centered.

**EPB isometry notes.** The isometry is exact and global: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to  $\rho$ ; unlike fixed random-feature anchors, whose pairwise distances grow nonlinearly and saturate, the isometric axis preserves label distances at every scale (with the price that anchor norms grow with  $|\tilde{y}|$ ; a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold). All prior means lie on one line; for  $y_i < y_j < y_k$  the distances add,  $\|\mu_p(y_i) - \mu_p(y_k)\| = \|\mu_p(y_i) - \mu_p(y_j)\| + \|\mu_p(y_j) - \mu_p(y_k)\|$ : order and magnitude of label differences are encoded exactly.

**Multidimensional regression labels.** For  $y \in \mathbb{R}^m$  ( $m \leq d$ ), the same construction extends verbatim: take  $W \in \mathbb{R}^{d \times m}$ , its thin polar factor  $Q = W(W^\top W)^{-1/2}$ , and the prior  $\mu_p(y) = \rho Q \tilde{y}$ , which is isometric in the label metric; the scalar case is  $m = 1$  with  $Q = u$ . The label metric must be declared before standardization, otherwise “isometric” has no unique meaning.

#### A.4 ADDITIONAL RESULTS

**Lemma A.1** (OPB KL decomposition). *For a diagonal posterior covariance and the prior of Eq. equation 9,*

$$\text{KL}(q_\phi(z|x) \| p_{\theta}^{\text{OPB}}(z|y)) = \frac{1}{2\tau^2} \|\mu_\phi(x) - a q_{\theta,y}\|^2 + \frac{1}{2} \sum_{r=1}^d \left[ \frac{\sigma_{\phi,r}^2(x)}{\tau^2} - 1 - \log \frac{\sigma_{\phi,r}^2(x)}{\tau^2} \right], \quad (39)$$

*a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at  $\sigma_{\phi,r}^2(x) = \tau^2$  for all  $r$ . The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition A.4 below.*

*Proof.* Apply the diagonal-Gaussian KL formula with posterior mean  $\mu_\phi(x)$ , prior mean  $a q_{\theta,y}$ , posterior variances  $\sigma_{\phi,r}^2(x)$ , and the fixed prior variance  $\tau^2$ .  $\square$

**Corollary A.1** (Label-information lower bound in the aligned Gaussian model). *Assume uniform classes and exact alignment  $q(z|Y = k) = \mathcal{N}(a q_{\theta,k}, \tau^2 I_d)$ . The pairwise error probability of the nearest-anchor classifier is  $P_{\text{pair}} = \Phi(-\frac{a}{\sqrt{2}\tau})$ , with  $P_e \leq \min\{1, (K-1)P_{\text{pair}}\}$ , and whenever the displayed bound  $\bar{P}_e \leq 1 - 1/K$ , Fano’s inequality gives  $I(Y; Z) \geq \log K - h_2(\bar{P}_e) - \bar{P}_e \log(K-1)$ . This links the anchor scale-to-noise ratio  $a/\tau$  to the retained label information, conditionally on the aligned Gaussian model.*

*Proof.* Two anchors have distance  $\sqrt{2}a$  by Proposition A.3; the equal-covariance binary decision boundary is halfway between them, giving  $\Phi(-\sqrt{2}a/(2\tau))$ . The union bound and Fano’s inequality on  $[0, 1 - 1/K]$  conclude.  $\square$

**Proposition A.4** (Joint strict convexity and unique optimum on a restricted aligned surrogate). *Consider the class-symmetric aligned family: every input of class  $y$  is mapped to the posterior  $q_y = \mathcal{N}(s q_{\theta,y}, \sigma^2 I)$  ( $s \geq 0$ ,  $\sigma^2 > 0$ ; anchor scale  $a > 0$ ), and the classifier is the anchor-energy classifier of Proposition 2.2 (weights  $w_j = c q_{\theta,j}$  with scale  $c = a/\tau^2$  fixed by construction). With the stochastic training code  $z \sim q_y$  (as in the actual objective, Eq. equation 11), the training cross-entropy is*

$$g(s, \sigma^2) := \mathbb{E}_\eta \left[ \ln \left( 1 + \sum_{j \neq y} e^{-cs + c\sigma(\eta_j - \eta_y)} \right) \right], \quad \eta \sim \mathcal{N}(0, I_K), \quad (40)$$

and the training Lagrangian is

$$L_\beta(s, \sigma^2) = g(s, \sigma^2) + \beta \left[ \frac{1}{2\tau^2}(s - a)^2 + \frac{d}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)) \right]. \quad (41)$$

Then for every  $\beta > 0$ ,  $L_\beta$  is strictly convex jointly in  $(s, \sigma)$  and has a unique joint minimizer  $(s^*(\beta), \sigma^*(\beta))$ , interior in  $s$  since  $a > 0$ : the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every  $\beta$  (Proposition A.1). This is a restricted surrogate result: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective, with its nonlinear posterior, trainable frame, and non-convex degrees of freedom, lies outside this analysis, and the result neither establishes  $\beta$ -tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code ( $z = \mu$ ) is the  $\sigma \rightarrow 0$  limit, and  $g \geq \ln(1 + (K - 1)e^{-cs})$  by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do not claim here that the map  $\beta \mapsto s^*(\beta)$  is monotone or that the sweep is one-to-one: the minimizer  $\sigma^*$  responds to  $\beta$  through the cross-derivative  $\partial^2 g / \partial s \partial \sigma$ , and establishing the sign of  $ds^* / d\beta$  requires the full two-dimensional stationarity analysis, which we leave open; the  $\beta$ -sweep is reported empirically in the experiments. The fixed-scale condition holds for the main model by construction (Proposition 2.2), so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution: at  $KL \rightarrow 0$  the posterior variance is pinned at  $\sigma^2 = \tau^2 > 0$ , so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for  $K = 2$  it contains  $\mathbb{E}[\ln(1 + e^{-c(a+\tau\xi)})]$ , whose integrand grows linearly in  $c$  on the event  $\xi < -a/\tau$ , which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.

*Proof.* For each fixed  $\eta$ , the integrand of Eq. equation 40 is a log-sum-exp of functions affine in  $(s, \sigma)$  (the constant 1 has slope zero and each exponential has slope  $(-c, c(\eta_j - \eta_y))$ ), hence jointly convex in  $(s, \sigma)$ , and the expectation  $g$  is jointly convex. No strictness is claimed for  $g$ : for  $K = 2$  the integrand depends on  $(s, \sigma)$  only through the single linear combination  $-cs + c\sigma(\eta_2 - \eta_1)$ , so its Hessian has rank one. Write the KL term as  $R(s, \sigma) := \frac{1}{2\tau^2}(s - a)^2 + \frac{d}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2))$ ; its Hessian in  $(s, \sigma)$  is  $\nabla^2 R = \text{diag}(\tau^{-2}, d(\tau^{-2} + \sigma^{-2}))$ , strictly positive definite for every  $\sigma > 0$ , so  $\beta R$  is strictly convex in  $(s, \sigma)$ . Hence  $L_\beta = g + \beta R$  is the sum of a jointly convex and a strictly convex function, and is strictly convex in  $(s, \sigma)$ . Since  $R \rightarrow +\infty$  as  $s \rightarrow \infty$ ,  $\sigma \rightarrow 0^+$ , or  $\sigma \rightarrow \infty$ , the sublevel sets of  $L_\beta$  are compact, and the unique joint minimizer exists (by strict convexity it is unique). For  $a > 0$  the minimizer is interior in  $s$ : at  $s = 0$ ,  $\partial L_\beta / \partial s = -c\mathbb{E}[p_w] - \beta a/\tau^2 < 0$ , where  $p_w$  is the error probability of the sampled code, so  $s^* > 0$ .  $\square$

*Remark A.4* (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition A.4 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale: a restricted surrogate result, which does not establish  $\beta$ -tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB’s: Eq. equation 14 is the usual CEB gap identity for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on Assumption 2.1: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified, not imposed, along the reported runs (Section 2.2.1); the construction is exactly equivariant under relabeling of the classes by Eq. equation 38, so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and Proposition 2.3 would no longer follow directly. Finally, the anchor-energy classifier of Proposition 2.2 fixes the classifier scale to  $a/\tau^2$  by construction, removing the independent classifier-margin escape route in the aligned surrogate of Proposition A.4; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition A.4

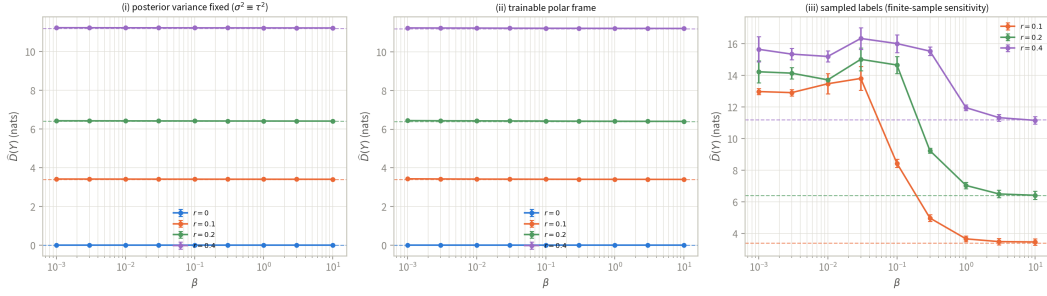


Figure 9: Sensitivity analyses for the controlled label-ambiguity experiment. (a) Posterior variance fixed to the prior variance. (b) Trainable stateless polar frame. (c) Labels sampled rather than integrated with their exact population weights; failed comparator-gate configurations are marked. Dashed lines denote the analytic population ambiguity floors. Points are means and error bars are standard deviations over five seeds.

moves monotonically in  $\beta$  is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to  $\beta$  through the cross-derivative  $\partial^2 g / \partial s \partial \sigma$  of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate’s zero set contains the full-retention orthogonal label codes at any frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code*; it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

#### A.5 CONTROLLED LABEL-AMBIGUITY SENSITIVITY ANALYSES

The population-weighted experiment of Section 3.2 uses a fixed frame and the standard trainable posterior variance. Figure 9 changes these choices one at a time. Fixing the posterior variance to  $\sigma^2 \equiv \tau^2$  leaves the measured mismatch on the analytic floors, confirming that the result is not produced by a covariance compensation channel. Replacing the fixed frame by the trainable stateless polar frame gives the same result, so the floor is invariant to the learned rigid frame rotation.

The third panel replaces the exact population weights by sampled noisy labels. It shows the expected finite-sample displacement from the population floor at weak compression and convergence toward the empirical floor as compression strengthens. Six sampled-label configurations at large  $\beta$  fail the explicit comparator-objective gate (twelve stored rows because both best and final checkpoints are retained); they are marked as optimization failures rather than interpreted as violations of Eq. equation 21. The population-weighted main grid and both structural controls satisfy the gate at every point.

#### A.6 DIRECT MISMATCH DEFINITIONS AND SEPARATION DIAGNOSTICS

This section gives the finite-sample quantities behind Section 3.8.1. All statistics are first computed within each seed and then aggregated across the five seeds; class or bin pairs are not treated as independent replicates. Every checkpoint in both requested grids is present, and no collapse or variance-explosion point is removed.

**Classification.** For test examples of class  $k$ , define

$$\hat{D}_k := \frac{1}{n_k} \sum_{n: y_n = k} \frac{\|\mu_\phi(x_n) - aq_{\theta,k}\|^2}{2\hat{\tau}_{\max}^2}, \quad \hat{m}_k := \frac{1}{n_k} \sum_{n: y_n = k} \mu_\phi(x_n). \quad (42)$$

Jensen gives the per-class check  $\|\hat{m}_k - aq_{\theta,k}\| \leq \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_k}$ , and the triangle inequality gives Eq. equation 37. Both are evaluated without clipping the lower bound. Across all OPB scales,  $\beta$  values, seeds, and classes, no Jensen or separation-slack violation exceeds  $10^{-6}$ . Figure 10 shows

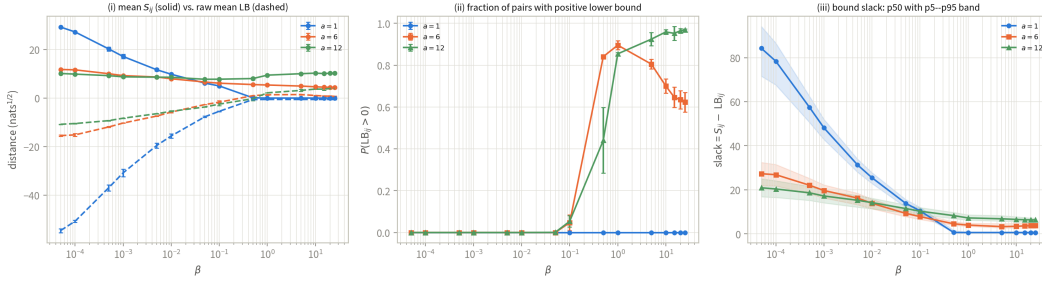


Figure 10: ImageNet-100 class-separation diagnostics. (a) Mean posterior-center distance and the raw (unclipped) mismatch-dependent lower bound against  $\beta$ ; CEB mismatch is included only as a descriptive reference. (b) Fraction of OPB class pairs with a positive lower bound. (c) The 5th, 50th, and 95th percentiles of the separation slack. Statistics are computed within seed before aggregation.

how the actual separation, raw lower bound, fraction of informative pairs, and slack change over the sweep. CEB’s mismatch is shown descriptively, but its bound and slack entries are left empty because its reference has no declared  $\Delta$ .

**Regression.** We partition the standardized test labels into 20 equal-width bins and merge any bin containing fewer than 30 examples. For bin  $B_b$ , let  $\bar{y}_b$  be its mean label,  $\hat{m}_b$  its posterior center,  $\hat{D}_b$  its average sample-to-own-anchor mismatch, and  $h_b := \text{diam}(B_b)$ . The finite-bin lower bound used in the diagnostic is

$$\widehat{\text{LB}}_{bc} = \rho|\bar{y}_b - \bar{y}_c| - \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_b} - \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_c} - \rho(h_b + h_c). \quad (43)$$

The last term is the quantization correction from Remark A.2. Figure 11 plots the actual bin-center distances against the declared isometric distances for four  $\beta$  values at  $\rho = 6$ . The points lie systematically below the identity line, consistent with the measured normalized axial slope of 0.73–0.77. After the quantization correction, 43% of bin pairs have a positive bound; all raw slacks remain non-negative to numerical tolerance.

**Normalization and variance extremes.** The empirical mismatch uses one maximum conditional-reference variance per checkpoint, whereas the exact Gaussian KL uses every sample’s and coordinate’s own variance. They therefore need not coincide: averaged over the ImageNet-100 grid,  $\hat{D} = 41.2$ , the KL mean term is 46.5, and the KL variance term is 51.4. On Housing, a small number of weak-compression, large- $\rho$  variance explosions raise the across-grid mean KL variance term to  $2.8 \times 10^6$ . These checkpoints remain in all plots. Because their  $\hat{\tau}_{\max}^2$  is also large, the normalized  $\hat{D}$  can fall even as the KL variance term grows; the unnormalized axial/off-axis components and total KL are therefore reported alongside it.

## A.7 MAIN RESULTS WITH STANDARD DEVIATIONS

Tables 7 and 8 report the same configurations as the main table with standard deviations over runs, split by column groups for width (the baselines table includes NCBD and the variants table includes AdaCap); bold and underline follow the same mean-based rule as Table 2.

## A.8 PAIRED-DIFFERENCE CONFIDENCE INTERVALS

Tables 9–12 report, for each setting, the paired difference of GPB over every baseline (GPB minus baseline) with a bootstrap 95% confidence interval: the differences are paired across the shared seeds, the interval is the percentile interval of  $B = 10,000$  resamples of the per-run differences (fixed seed, reproducible), and the non-inferiority margin is  $\delta = 0.2$  percentage points of accuracy or  $\delta = 0.005$  in  $R^2$ . Bold marks a lower bound above zero (significant advantage) and  $\dagger$  a lower bound within the margin (tied); the best configuration per method is the same as in the main table. The last table covers the external non-IB baselines, NCBD on the classification settings and AdaCap

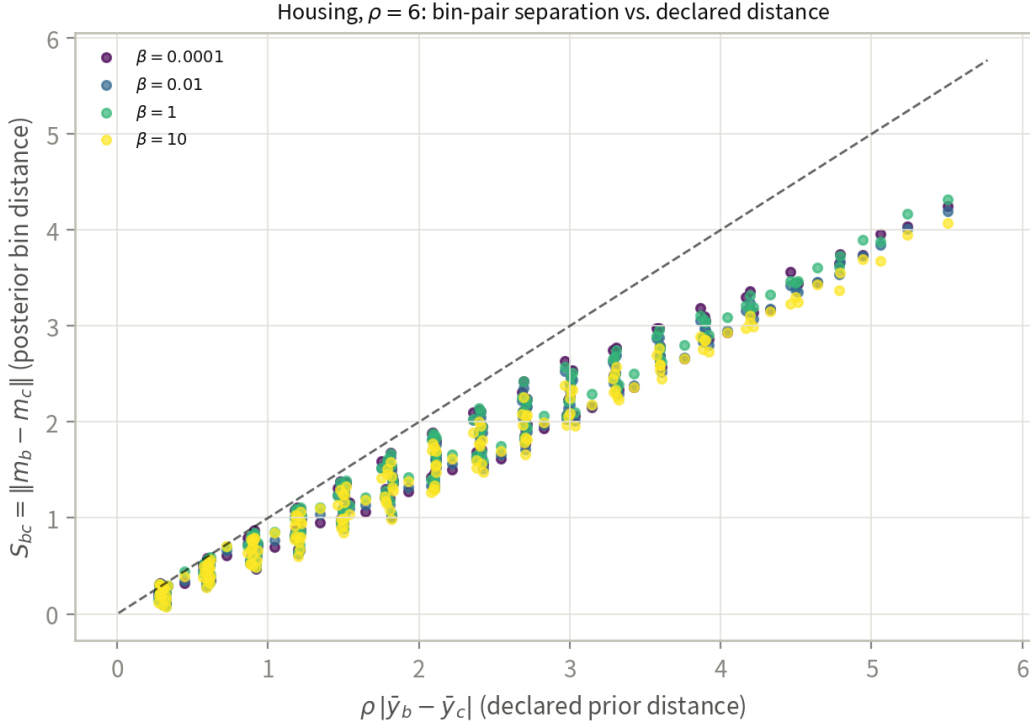


Figure 11: Cal. Housing finite-bin regression diagnostic at  $\rho = 6$ . Each panel compares posterior bin-center distance with the declared prior distance  $\rho|\bar{y}_b - \bar{y}_c|$  at one  $\beta$ ; the identity line denotes exact scale transfer. The mismatch-dependent bound additionally includes the two mismatch terms and the bin-diameter correction of Eq. equation 43.

Table 7: Test accuracy (%) and test  $R^2$  with standard deviations over runs: the plain backbone, the variational baselines, and NCBD (classification settings only); same configurations as Table 2.

| Dataset      | Backbone | Base                                | VIB                 | SVIB                | NIB                 | NCBD             |
|--------------|----------|-------------------------------------|---------------------|---------------------|---------------------|------------------|
| MNIST        | MLP      | 98.33 $\pm$ 0.12                    | 98.28 $\pm$ 0.11    | 98.15 $\pm$ 0.13    | 98.27 $\pm$ 0.07    | 97.95 $\pm$ 0.23 |
| MNIST        | CNN      | 99.13 $\pm$ 0.05                    | 99.28 $\pm$ 0.09    | 99.23 $\pm$ 0.03    | 99.22 $\pm$ 0.06    | 99.18 $\pm$ 0.07 |
| ImageNet-100 | MLP      | 83.12 $\pm$ 0.14                    | 82.84 $\pm$ 0.44    | 83.02 $\pm$ 0.37    | 82.35 $\pm$ 0.27    | 83.64 $\pm$ 0.07 |
| ImageNet-100 | CNN      | 83.40 $\pm$ 0.13                    | 82.94 $\pm$ 0.37    | 83.11 $\pm$ 0.86    | 82.91 $\pm$ 0.35    | 80.56 $\pm$ 1.86 |
| Cora         | GCN      | 87.68 $\pm$ 0.46                    | 86.83 $\pm$ 0.28    | 86.46 $\pm$ 1.87    | 86.35 $\pm$ 0.68    | 87.34 $\pm$ 1.60 |
| IMDb         | RNN      | 84.85 $\pm$ 0.68                    | 85.50 $\pm$ 0.91    | 85.99 $\pm$ 0.30    | 86.14 $\pm$ 0.92    | 86.69 $\pm$ 0.52 |
| AG News      | RNN      | <b>92.94<math>\pm</math>0.15</b>    | 92.72 $\pm$ 0.17    | 92.86 $\pm$ 0.21    | 92.91 $\pm$ 0.22    | 92.37 $\pm$ 0.37 |
| Cal. Housing | MLP      | 0.7618 $\pm$ 0.0238                 | 0.7901 $\pm$ 0.0049 | 0.7897 $\pm$ 0.0058 | 0.7862 $\pm$ 0.0035 | –                |
| STS-B        | RNN      | <b>0.2899<math>\pm</math>0.0113</b> | 0.2719 $\pm$ 0.0164 | 0.2697 $\pm$ 0.0202 | 0.2417 $\pm$ 0.0221 | –                |
| ZINC         | GCN      | 0.6348 $\pm$ 0.0044                 | 0.6339 $\pm$ 0.0033 | 0.6325 $\pm$ 0.0026 | 0.6178 $\pm$ 0.0099 | –                |
| AgeDB        | MLP      | 0.2180 $\pm$ 0.0409                 | 0.3173 $\pm$ 0.0101 | 0.3385 $\pm$ 0.0286 | 0.2710 $\pm$ 0.0235 | –                |
| AgeDB        | CNN      | 0.3914 $\pm$ 0.0194                 | 0.3342 $\pm$ 0.1894 | 0.3323 $\pm$ 0.1870 | 0.3337 $\pm$ 0.1889 | –                |

on the regression settings; the only interval lying entirely below zero is AdaCap on ZINC ( $-0.043 R^2$ ).

#### A.9 INFORMATION-PLANE PANELS FOR CAL. HOUSING

Figure 12 completes Fig. 3 of the main text with the regression setting, under the same  $\beta$  grid, axis conventions, and five runs per point (Section 3.6); here  $I(Y; Z)$  is the Gaussian approximation  $\frac{1}{2} \ln(\text{Var}(y)/\text{Var}(y - \hat{y}))$ , a heuristic rather than a bound. On Cal. Housing this approximation consistently reads above the InfoNCE bound, so the certificate panel reflects the estimator gap rather

Table 8: Test accuracy (%) and test  $R^2$  with standard deviations over runs: CEB, DVCCA, AdaCap (regression settings only), and our method; same configurations as Table 2. GPB and GPB-L are separated from the references by the vertical rule.

| Dataset      | Backbone | CEB           | DVCCA         | AdaCap               | GPB                  | GPB-L                |
|--------------|----------|---------------|---------------|----------------------|----------------------|----------------------|
| MNIST        | MLP      | 98.43±0.08    | 98.24±0.17    | –                    | <b>98.54±0.09</b>    | 98.43±0.07           |
| MNIST        | CNN      | 99.30±0.05    | 99.29±0.10    | –                    | <b>99.32±0.05</b>    | 99.31±0.05           |
| ImageNet-100 | MLP      | 82.77±0.22    | 82.99±0.43    | –                    | <b>84.40±0.21</b>    | 84.01±0.23           |
| ImageNet-100 | CNN      | 82.97±0.67    | 83.21±0.70    | –                    | <b>86.08±0.21</b>    | 84.88±0.80           |
| Cora         | GCN      | 86.27±0.38    | 86.90±0.94    | –                    | <b>88.12±0.55</b>    | 87.86±0.87           |
| IMDb         | RNN      | 85.55±0.73    | 85.69±0.23    | –                    | 86.84±0.57           | <b>86.99±0.35</b>    |
| AG News      | RNN      | 92.83±0.11    | 92.78±0.15    | –                    | 92.87±0.12           | 92.92±0.14           |
| Cal. Housing | MLP      | 0.7894±0.0018 | 0.7878±0.0048 | 0.7758±0.0023        | 0.7895±0.0051        | <b>0.7914±0.0034</b> |
| STS-B        | RNN      | 0.2668±0.0127 | 0.2713±0.0218 | 0.1613±0.0188        | 0.2762±0.0152        | 0.2755±0.0134        |
| ZINC         | GCN      | 0.6257±0.0044 | 0.6294±0.0026 | <b>0.6770±0.0019</b> | 0.6344±0.0031        | 0.6323±0.0016        |
| AgeDB        | MLP      | 0.3317±0.0251 | 0.3180±0.0228 | 0.3482±0.0209        | <b>0.3536±0.0076</b> | 0.3320±0.0080        |
| AgeDB        | CNN      | 0.3882±0.0275 | 0.3205±0.1811 | 0.4684±0.0094        | <b>0.4700±0.0072</b> | 0.4388±0.0220        |

Table 9: Paired-difference 95% confidence intervals of GPB over the plain backbone and VIB (GPB minus baseline, per setting; classification in percentage points, regression in  $R^2$ ). Bold: lower CI bound above zero; †: lower bound within the non-inferiority margin  $\delta$  (tied).

| Dataset      | Backbone | Base                           | VIB                            |
|--------------|----------|--------------------------------|--------------------------------|
| MNIST        | MLP      | <b>+0.2 [+0.2, +0.3]</b>       | <b>+0.3 [+0.2, +0.3]</b>       |
| MNIST        | CNN      | <b>+0.2 [+0.2, +0.2]</b>       | +0.0 [-0.0, +0.1]†             |
| ImageNet-100 | MLP      | <b>+1.3 [+1.2, +1.4]</b>       | <b>+1.6 [+1.4, +1.9]</b>       |
| ImageNet-100 | CNN      | <b>+2.7 [+2.5, +2.9]</b>       | <b>+3.1 [+3.0, +3.4]</b>       |
| Cora         | GCN      | +0.4 [-0.1, +1.1]†             | <b>+1.3 [+0.7, +1.7]</b>       |
| IMDb         | RNN      | <b>+2.0 [+1.2, +2.7]</b>       | <b>+1.3 [+0.6, +2.3]</b>       |
| AG News      | RNN      | -0.1 [-0.2, +0.0]†             | <b>+0.2 [+0.1, +0.2]</b>       |
| Cal. Housing | MLP      | <b>+0.028 [+0.008, +0.048]</b> | -0.001 [-0.005, +0.003]†       |
| STS-B        | RNN      | -0.014 [-0.022, -0.006]        | +0.004 [-0.017, +0.022]        |
| ZINC         | GCN      | -0.000 [-0.003, +0.002]†       | +0.001 [-0.004, +0.004]†       |
| AgeDB        | MLP      | <b>+0.136 [+0.104, +0.167]</b> | <b>+0.036 [+0.026, +0.046]</b> |
| AgeDB        | CNN      | <b>+0.079 [+0.066, +0.091]</b> | <b>+0.136 [+0.043, +0.302]</b> |
|              |          | 8*/3†                          | 8*/3†                          |

than the geometry; the information plane remains informative: CEB compresses itself off the task ( $R^2 = 0.49$  at  $\beta = 0.1$ ,  $0.18$  at  $\beta = 0.5$ ) while every EPB curve keeps  $R^2 \geq 0.76$  flat out to  $\beta = 25$ .

Table 10: Paired-difference 95% confidence intervals of GPB over SVIB and NIB (GPB minus baseline, per setting; classification in percentage points, regression in  $R^2$ ). Bold/ $\dagger$  as in the baselines table.

| Dataset      | Backbone | SVIB                                            | NIB                                             |
|--------------|----------|-------------------------------------------------|-------------------------------------------------|
| MNIST        | MLP      | <b>+0.4</b> [ <b>+0.3</b> , <b>+0.5</b> ]       | <b>+0.3</b> [ <b>+0.2</b> , <b>+0.3</b> ]       |
| MNIST        | CNN      | <b>+0.1</b> [ <b>+0.1</b> , <b>+0.1</b> ]       | <b>+0.1</b> [ <b>+0.1</b> , <b>+0.2</b> ]       |
| ImageNet-100 | MLP      | <b>+1.4</b> [ <b>+1.1</b> , <b>+1.7</b> ]       | <b>+2.1</b> [ <b>+1.8</b> , <b>+2.2</b> ]       |
| ImageNet-100 | CNN      | <b>+3.0</b> [ <b>+2.4</b> , <b>+3.9</b> ]       | <b>+3.2</b> [ <b>+2.9</b> , <b>+3.5</b> ]       |
| Cora         | GCN      | <b>+1.7</b> [ <b>+0.2</b> , <b>+3.4</b> ]       | <b>+1.8</b> [ <b>+1.1</b> , <b>+2.4</b> ]       |
| IMDb         | RNN      | <b>+0.8</b> [ <b>+0.2</b> , <b>+1.4</b> ]       | <b>+0.7</b> [ <b>+0.2</b> , <b>+1.2</b> ]       |
| AG News      | RNN      | +0.0 [-0.1, +0.1] $\dagger$                     | -0.0 [-0.2, +0.1] $\dagger$                     |
| Cal. Housing | MLP      | -0.000 [-0.006, +0.004]                         | +0.003 [-0.001, +0.009] $\dagger$               |
| STS-B        | RNN      | +0.006 [-0.021, +0.028]                         | <b>+0.035</b> [ <b>+0.011</b> , <b>+0.058</b> ] |
| ZINC         | GCN      | +0.002 [-0.001, +0.004] $\dagger$               | <b>+0.017</b> [ <b>+0.010</b> , <b>+0.026</b> ] |
| AgeDB        | MLP      | +0.015 [-0.010, +0.041]                         | <b>+0.083</b> [ <b>+0.062</b> , <b>+0.100</b> ] |
| AgeDB        | CNN      | <b>+0.138</b> [ <b>+0.045</b> , <b>+0.303</b> ] | <b>+0.136</b> [ <b>+0.040</b> , <b>+0.302</b> ] |
|              |          | 7*/2 $\dagger$                                  | 10*/2 $\dagger$                                 |

Table 11: Paired-difference 95% confidence intervals of GPB over CEB, DVCCA, and the free-head ablation GPB-L (GPB minus baseline, per setting; classification in percentage points, regression in  $R^2$ ). Bold/ $\dagger$  as in the baselines table.

| Dataset      | Backbone | CEB                                             | DVCCA                                           | GPB-L                                           |
|--------------|----------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| MNIST        | MLP      | <b>+0.1</b> [ <b>+0.0</b> , <b>+0.2</b> ]       | <b>+0.3</b> [ <b>+0.2</b> , <b>+0.4</b> ]       | <b>+0.1</b> [ <b>+0.0</b> , <b>+0.2</b> ]       |
| MNIST        | CNN      | +0.0 [-0.0, +0.1] $\dagger$                     | +0.0 [-0.1, +0.1] $\dagger$                     | +0.0 [-0.0, +0.1] $\dagger$                     |
| ImageNet-100 | MLP      | <b>+1.6</b> [ <b>+1.5</b> , <b>+1.7</b> ]       | <b>+1.4</b> [ <b>+1.1</b> , <b>+1.7</b> ]       | <b>+0.4</b> [ <b>+0.2</b> , <b>+0.5</b> ]       |
| ImageNet-100 | CNN      | <b>+3.1</b> [ <b>+2.4</b> , <b>+3.7</b> ]       | <b>+2.9</b> [ <b>+2.4</b> , <b>+3.5</b> ]       | <b>+1.2</b> [ <b>+0.5</b> , <b>+1.9</b> ]       |
| Cora         | GCN      | <b>+1.8</b> [ <b>+1.5</b> , <b>+2.3</b> ]       | <b>+1.2</b> [ <b>+0.4</b> , <b>+1.9</b> ]       | +0.3 [-0.5, +1.3]                               |
| IMDb         | RNN      | <b>+1.3</b> [ <b>+0.3</b> , <b>+2.1</b> ]       | <b>+1.1</b> [ <b>+0.8</b> , <b>+1.5</b> ]       | -0.2 [-0.6, +0.5]                               |
| AG News      | RNN      | +0.0 [-0.1, +0.2] $\dagger$                     | +0.1 [-0.0, +0.2] $\dagger$                     | -0.0 [-0.2, +0.1] $\dagger$                     |
| Cal. Housing | MLP      | +0.000 [-0.003, +0.004] $\dagger$               | +0.002 [-0.005, +0.009]                         | -0.002 [-0.005, +0.002]                         |
| STS-B        | RNN      | +0.009 [-0.009, +0.025]                         | +0.005 [-0.020, +0.026]                         | +0.001 [-0.019, +0.017]                         |
| ZINC         | GCN      | <b>+0.009</b> [ <b>+0.005</b> , <b>+0.011</b> ] | <b>+0.005</b> [ <b>+0.002</b> , <b>+0.007</b> ] | +0.002 [-0.001, +0.005] $\dagger$               |
| AgeDB        | MLP      | <b>+0.022</b> [ <b>+0.006</b> , <b>+0.038</b> ] | <b>+0.036</b> [ <b>+0.012</b> , <b>+0.057</b> ] | <b>+0.022</b> [ <b>+0.019</b> , <b>+0.025</b> ] |
| AgeDB        | CNN      | <b>+0.082</b> [ <b>+0.064</b> , <b>+0.099</b> ] | <b>+0.149</b> [ <b>+0.065</b> , <b>+0.310</b> ] | <b>+0.031</b> [ <b>+0.018</b> , <b>+0.046</b> ] |
|              |          | 8*/3 $\dagger$                                  | 8*/2 $\dagger$                                  | 5*/3 $\dagger$                                  |

Table 12: Paired-difference 95% confidence intervals of GPB over the external non-IB baselines NCBD (classification settings only) and AdaCap (regression settings only); GPB minus baseline, classification in percentage points, regression in  $R^2$ . Bold/ $\dagger$  as in the baselines table.

| Dataset      | Backbone | NCBD                                      | AdaCap                                          |
|--------------|----------|-------------------------------------------|-------------------------------------------------|
| MNIST        | MLP      | <b>+0.6</b> [ <b>+0.4</b> , <b>+0.8</b> ] | —                                               |
| MNIST        | CNN      | <b>+0.1</b> [ <b>+0.1</b> , <b>+0.2</b> ] | —                                               |
| ImageNet-100 | MLP      | <b>+0.8</b> [ <b>+0.6</b> , <b>+0.9</b> ] | —                                               |
| ImageNet-100 | CNN      | <b>+5.5</b> [ <b>+4.3</b> , <b>+6.9</b> ] | —                                               |
| Cora         | GCN      | <b>+0.8</b> [ <b>+0.0</b> , <b>+1.9</b> ] | —                                               |
| IMDb         | RNN      | +0.2 [-0.5, +0.7]                         | —                                               |
| AG News      | RNN      | <b>+0.5</b> [ <b>+0.2</b> , <b>+0.8</b> ] | —                                               |
| Cal. Housing | MLP      | —                                         | <b>+0.014</b> [ <b>+0.009</b> , <b>+0.019</b> ] |
| STS-B        | RNN      | —                                         | <b>+0.115</b> [ <b>+0.086</b> , <b>+0.135</b> ] |
| ZINC         | GCN      | —                                         | -0.043 [-0.046, -0.041]                         |
| AgeDB        | MLP      | —                                         | +0.005 [-0.007, +0.021]                         |
| AgeDB        | CNN      | —                                         | +0.002 [-0.004, +0.006] $\dagger$               |
|              |          | 6*/0 $\dagger$                            | 2*/1 $\dagger$                                  |



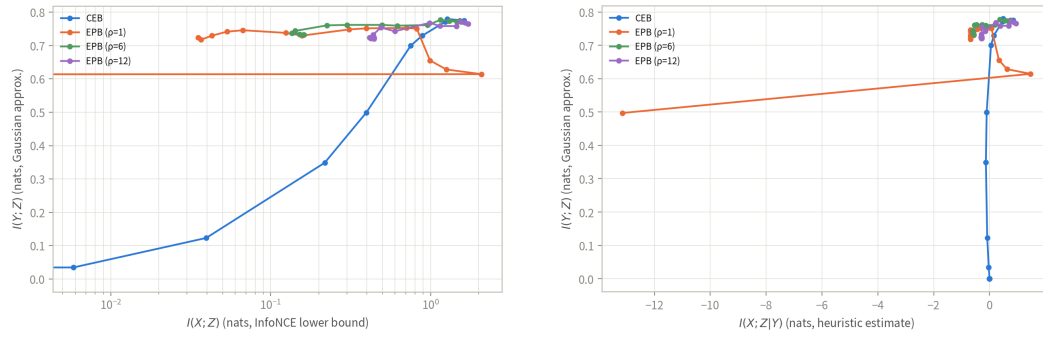


Figure 12: Information plane (a) and certificate plane (b) for Cal. Housing, companion panels to Fig. 3: identical estimators,  $\beta$  grid (log  $x$ -scale in (a); linear where the estimated residual turns negative in (b)), five runs per point, lines connect  $\beta$  in ascending order. CEB one curve per panel; EPB three ( $\rho \in \{1, 6, 12\}$ ).